Scanning Probe Microscopy (SPM)

"Seeing" at the nanoscale

- Atomic Force Microscopy (AFM)
- Scanning Tunneling Microscopy (STM)

History

- The Scanning Tunneling Microscope (STM) was invented by G. Binnig and H. Rohrer, for which they were awarded the Nobel Prize in 1986
- A few years later, the first Atomic Force Microscope (AFM) was developed by G. Binnig, Ch. Gerber, and C. Quate at Stanford University by gluing a tiny shard of diamond onto one end of a tiny strip of gold foil
- Currently AFM is the most common form of scanning probe microscopy

Inventors of STM

The Nobel Prize in Physics 1986

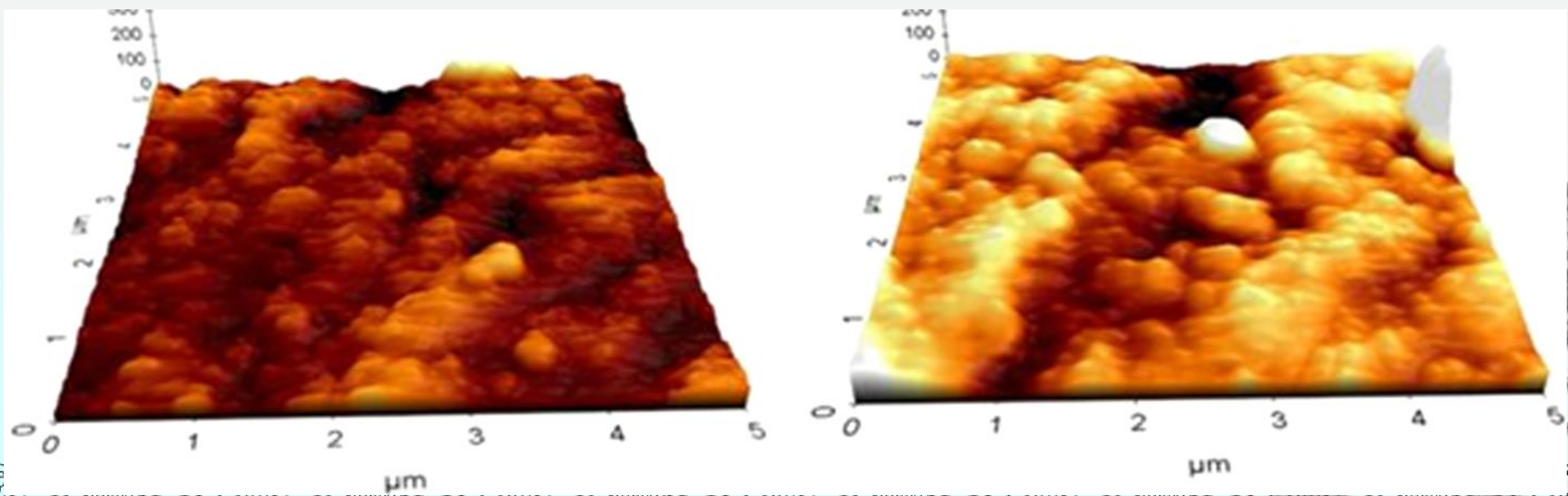
Nobel Laureates Heinrich Rohrer and Gerd Binnig

Atomic Force Microscopy

"Seeing" at the nanoscale

What does AFM do?

AFM provides a 3D profile of the surface on a nanoscale, by measuring forces between a sharp probe (<10nm) and surface at very short distance (0.2-10nm) probe-sample separation". The probe is supported on a flexible cantilever(悬臂). The AFM tip "gently" touches the surface and records the small force between the probe and the surface."

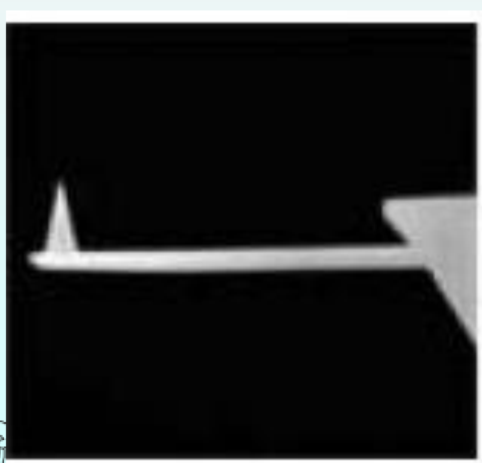


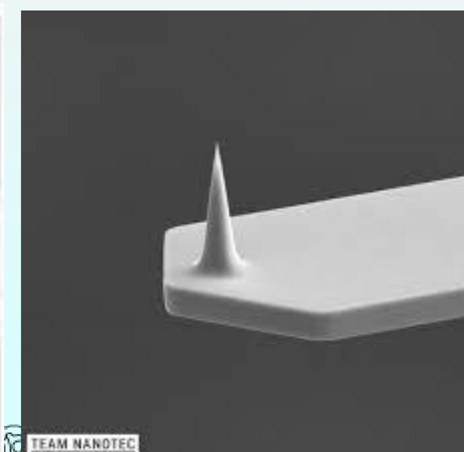
How does it measure the forces?

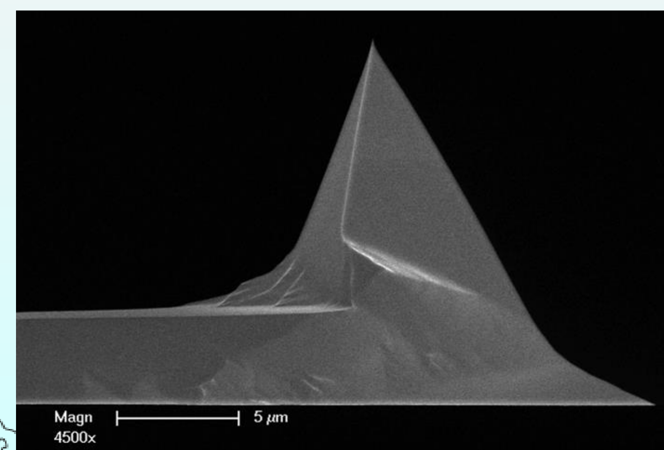
- The probe is placed on the end of a cantilever (which one can think of as a spring).
- The amount of force between the probe and sample is dependent on the spring constant (stiffness (刚度) of the cantilever, typical value are 0.1– 1N/m) and the distance between the probe and the sample surface.
- This force can be described using Hooke's Law

$$F=-k \bullet x$$

Typical forces are 10^{-6}-10^{-9} N

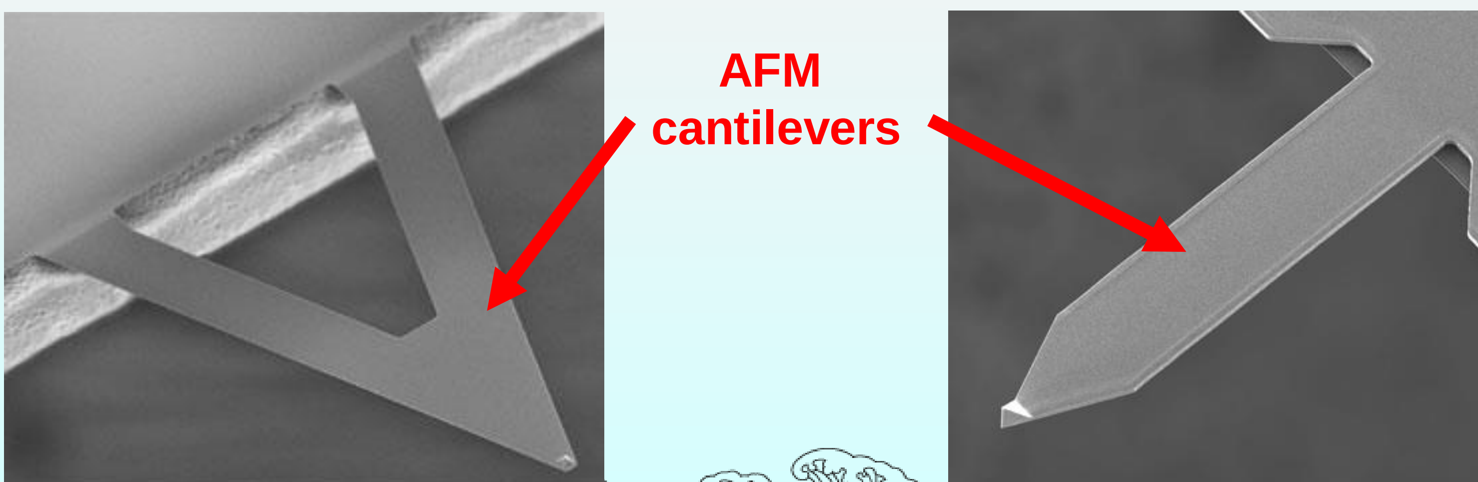






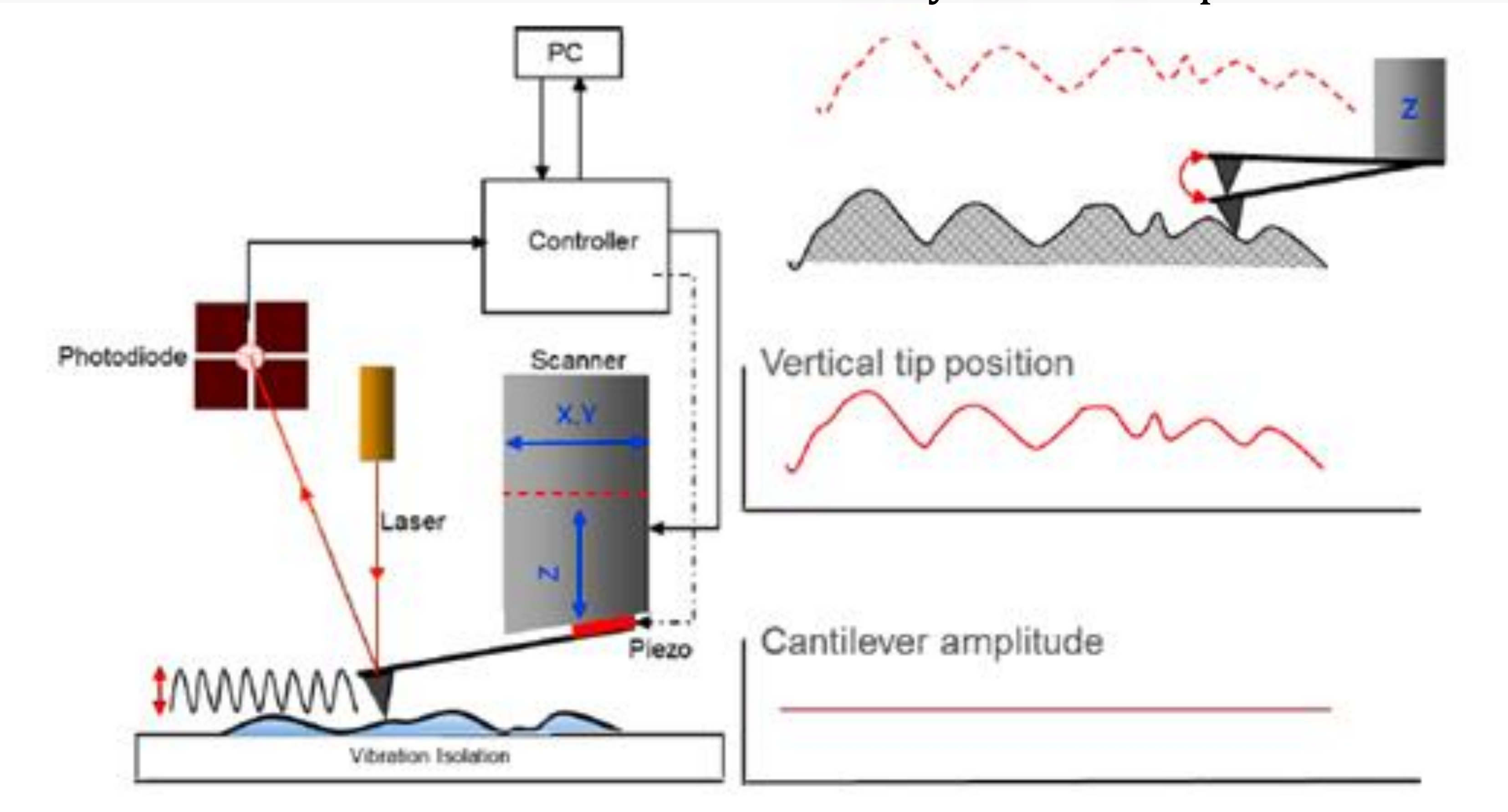
AFM cantilevers

- Probes are typically made from silicon nitride or silicon.
- Different cantilever lengths, materials, and shapes allow for varied spring constants and resonant frequencies.
- A description of the variety of different probes can be found on various vendor sites.
- Probes may be coated with other materials for additional scanning probe microscopy (SPM) applications such as chemical force microscopy (CFM) and magnetic force microscopy (MFM).”



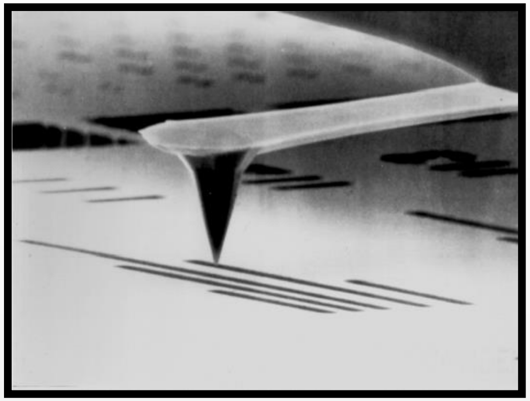
How a contact mode image is created

- Monitors the forces of attraction and repulsion between a probe and a sample surface, The cantilever is designed with a very low spring constant (easy to bend) so it is very sensitive to force.
- The tip is attached to a *cantilever* which moves up and down in response to forces of attraction or repulsion with the sample surface
  - Movement of the cantilever is detected by a laser and photodetector

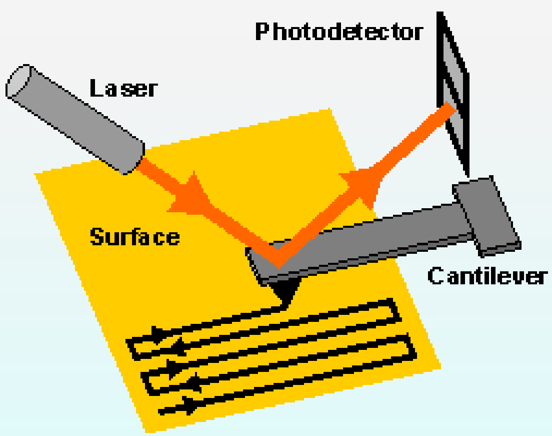


Raster the Tip: Generating an Image

Scanning Tip



Raster Motion



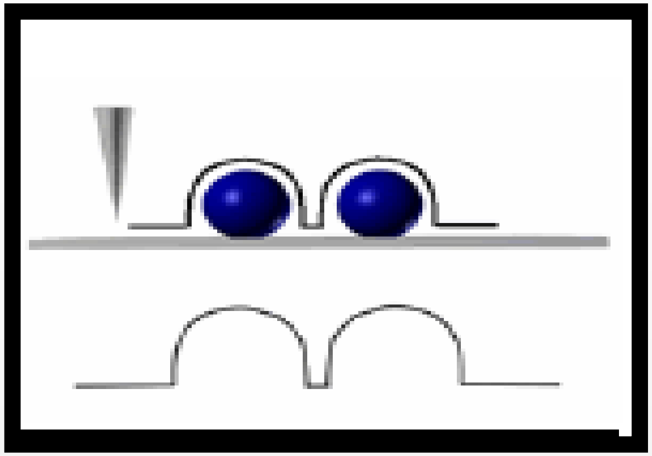
- The tip passes back and forth in a straight line across the sample (think old typewriter or CRT)
- In the typical imaging mode, the tip-sample force is held constant by adjusting the vertical position of the tip (feedback).
- A topographic image is built up by the computer by recording the vertical position as the tip is rastered across the sample.

Top Image Courtesy of Nanodevices, Inc. (www.nanodevices.com)

Bottom Image Courtesy of Stefanie Roes (www.fz-borstel.de/biophysik/_de/methods/afm.html)

Scanning the Sample

- Tip brought within nanometers of the sample (van der Waals)
- Radius of tip limits the accuracy of analysis/ resolution
- Stiffer cantilevers protect against sample damage because they deflect less in response to a small force
  - This means a more sensitive detection scheme is needed
  - measure change in resonance frequency and amplitude of oscillation



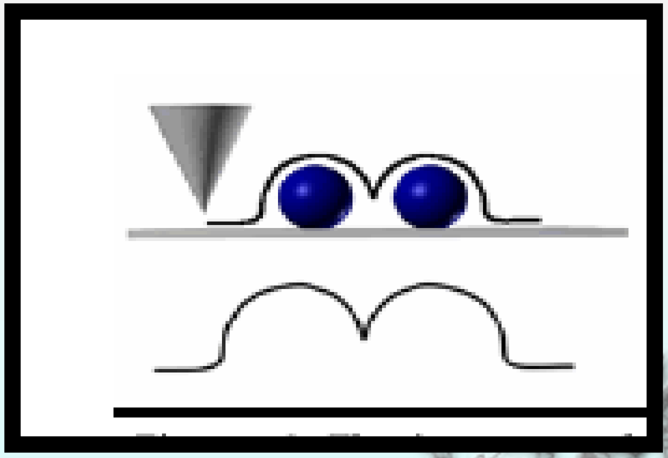
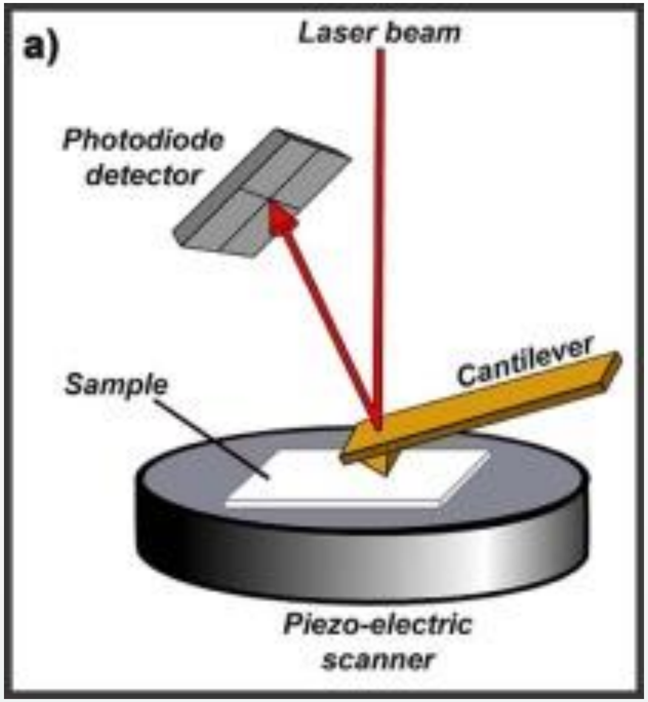
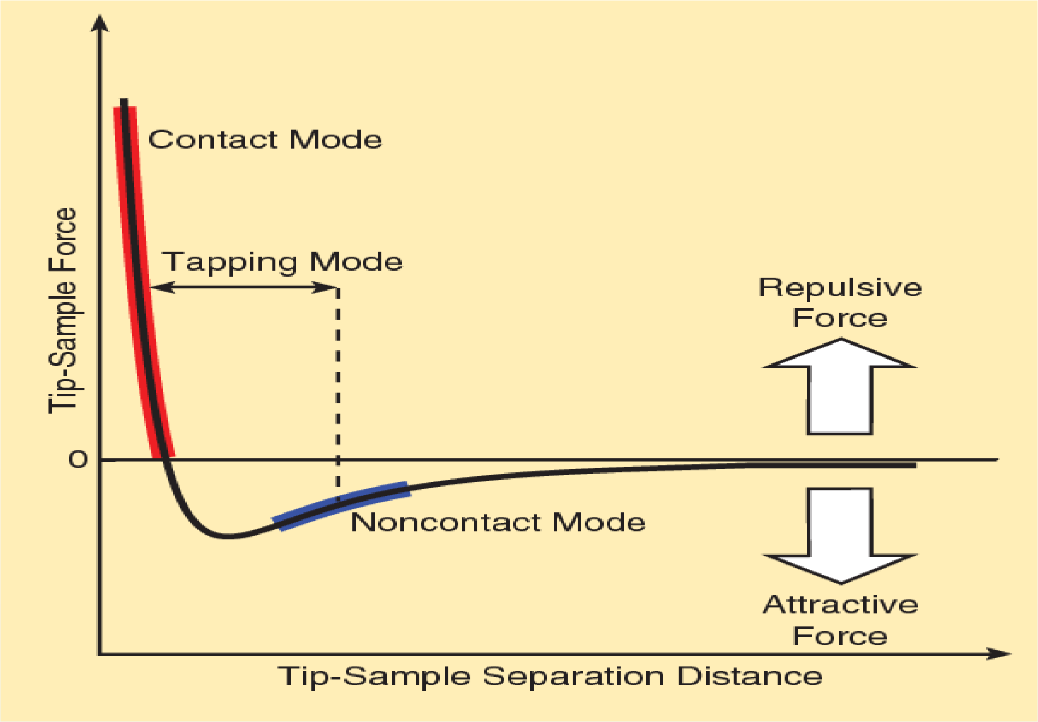


Image courtesy of (www.pacificnanotech.com)

Types of forces measured





"The dominant interactions at short probe-sample distances in the AFM are Van der Waals interactions. However, long-range interactions (i.e. capillary, electrostatic, magnetic) are significant further away from the surface. These are important in other SPM methods of analysis. During contact with the sample, the probe predominately experience *repulsive* Van der Waals forces (**contact mode**) to the tip deflection described previously. As the tip moves further away from the surface *attractive* Van der Waals forces are dominant (**non-contact mode.**)"

Modes of Operation

There are 3 primary imaging modes in AFM:

(1) Contact AFM

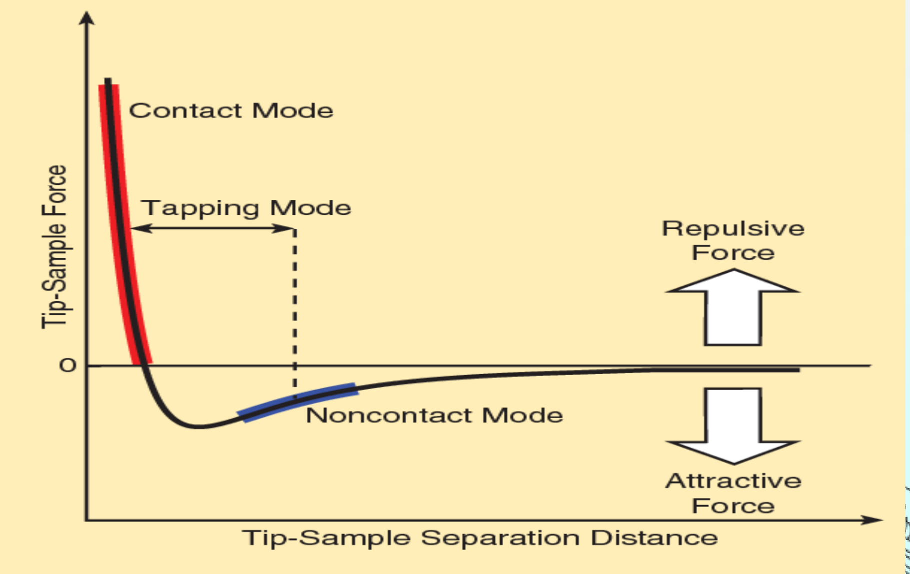
<0.5nm probe-surface separation, the forces are repulsive

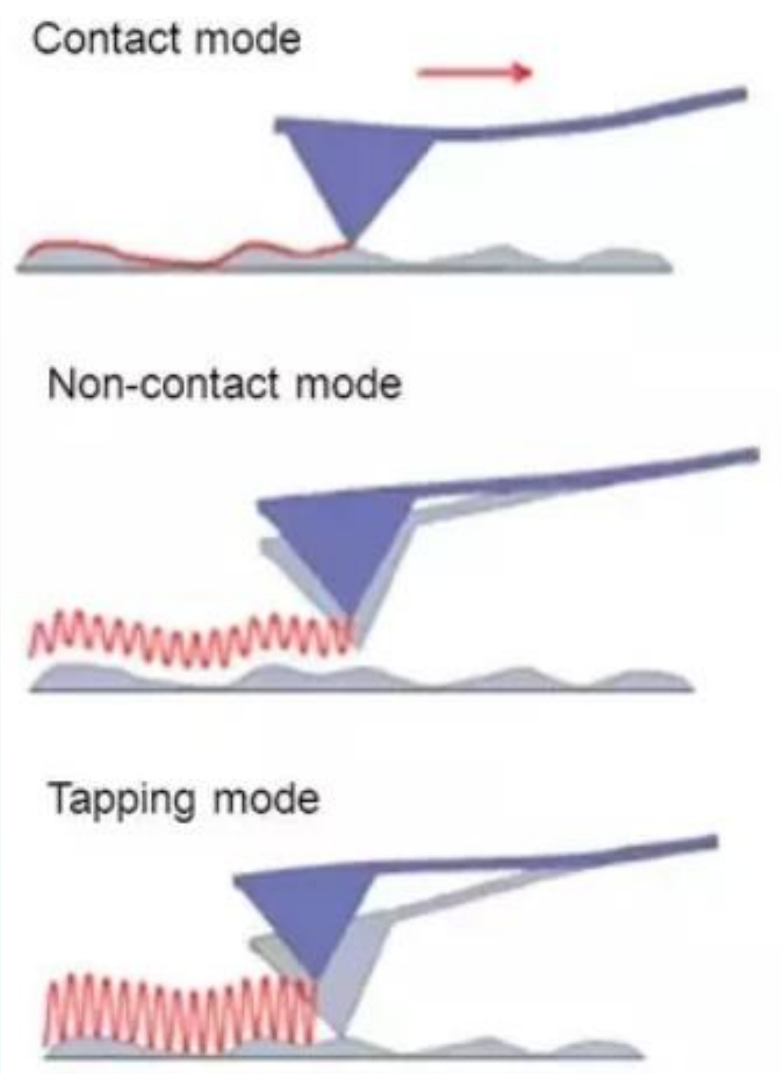
(2) Non-contact AFM

0.5-10 nm probe-surface separation, the forces are attractive

(3) Intermittent contact (tapping mode AFM)

0.5-2 nm probe-surface separation

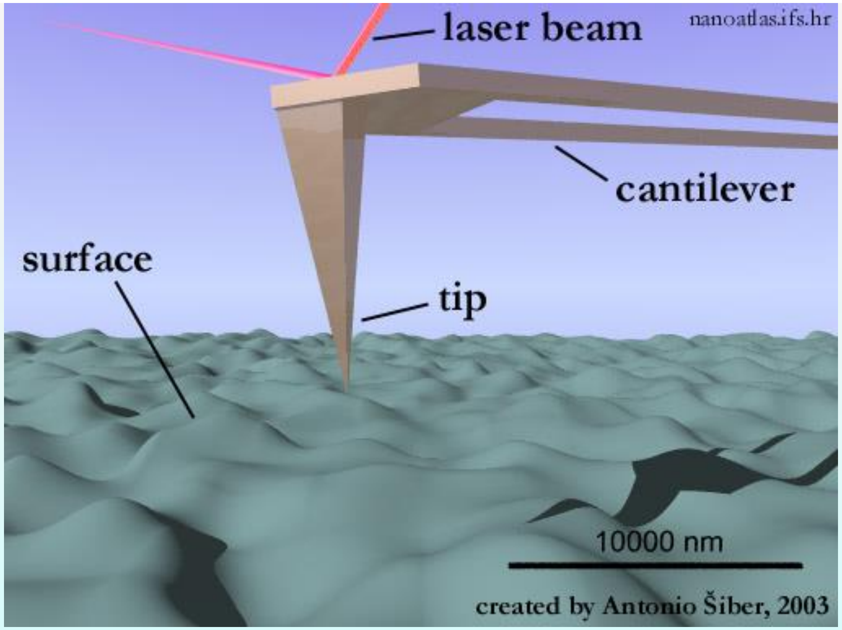




The AFM cantilever can be used to measure both attractive force mode and repulsive forces.

Contact Modes AFM

Contact Modes AFM: (repulsive Van der Waals) When the spring constant of cantilever is less than surface, the cantilever bends. The force on the tip is repulsive. By maintaining a constant cantilever deflection (using the feedback loops) the force between the probe and the sample remains constant and an image of the surface is obtained.

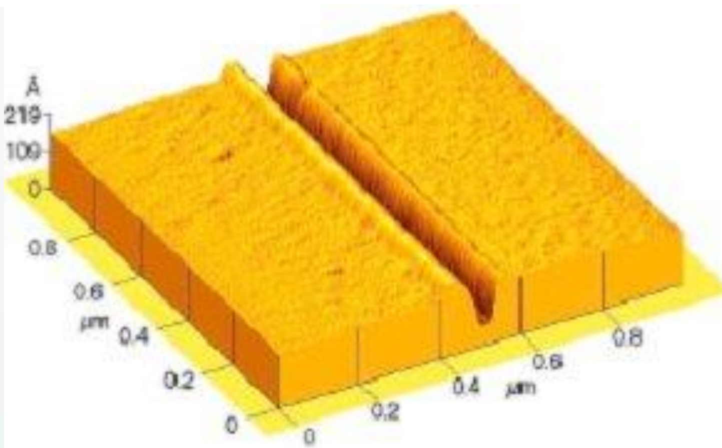


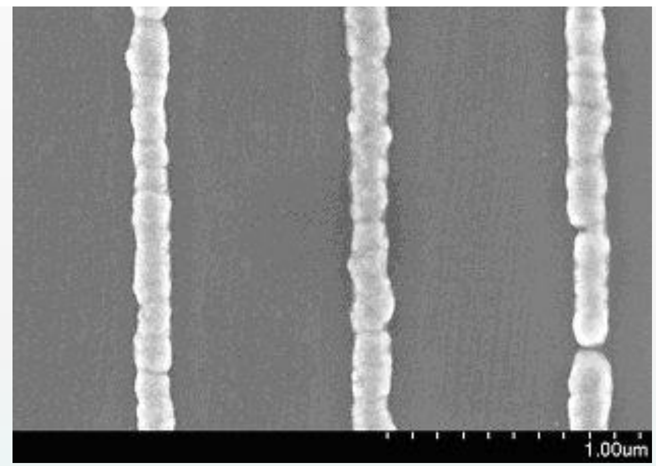
- **Advantages: fast scanning, good for rough samples, used in friction analysis.**
- **Disadvantages: at time force can damage/deform soft samples (however imaging in liquids often resolves this issue)**

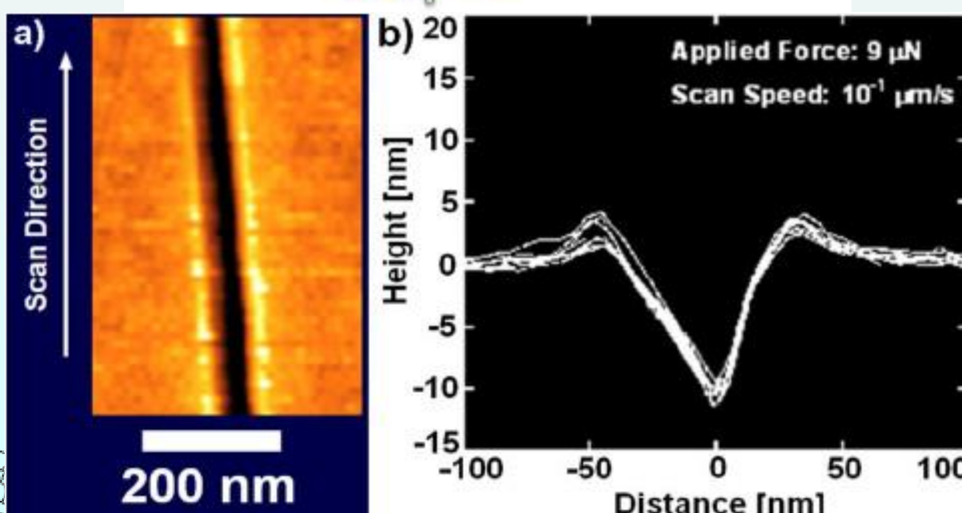
✓ In ambient conditions there is also a capillary force exerted by the thin water layer present (2-50 nm thick).

Scratching the Surface

If deformation of the surface is not desired, switch to a lower spring constant cantilever or switch to tapping or non-contact mode. **But sometimes, researchers want to nano-pattern the surface.**





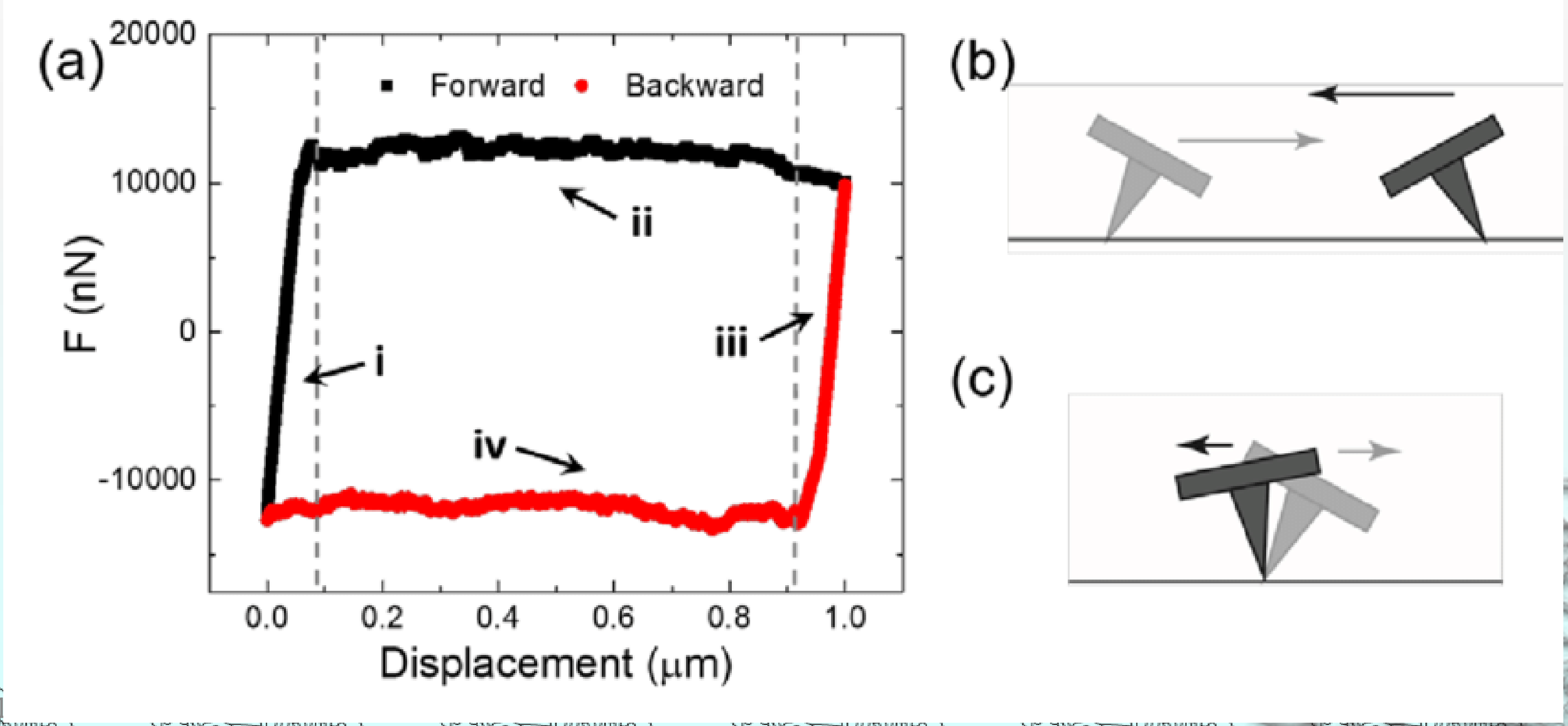


www.lko.uni-Erlangen.de/en/research/AFM.html

This research group introduces defects with AFM, and then electrodeposit metal. The metal deposits preferentially on the scratched sites due to electric field higher at smallest curvature.

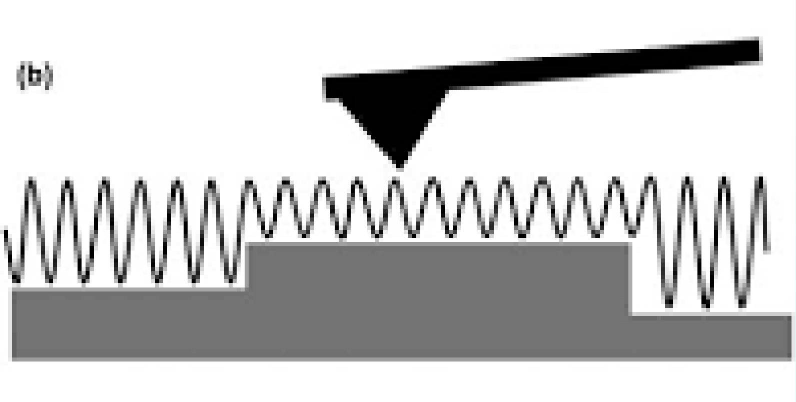
Measuring Friction on the Nanoscale

Lateral forces on the tip can cause the cantilever to bend sideways. This causes the deflection of the laser spot on the photodiode from left to right, as the tip slides back and forth. The degree of deflection can be used to measure friction on the nanoscale.



Tapping Mode

Intermittent Mode (Tapping): The imaging is similar to contact. However, in this mode the cantilever is oscillated at its resonant frequency, with oscillation amplitudes of 20-100 nm. The probe lightly "taps" on the sample surface during scanning, contacting the surface at the bottom of its swing. By maintaining a constant oscillation amplitude a constant tip-sample interaction is maintained and an image of the surface is obtained.



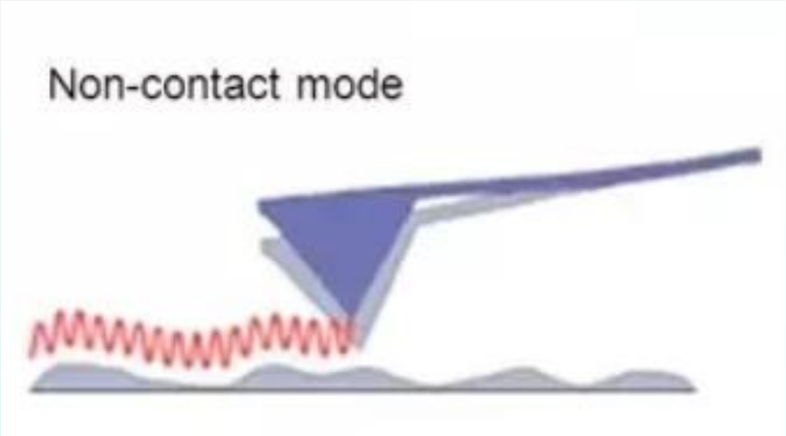
Tapping Mode

- **Advantages: allows high resolution of samples that easily damaged and/or loosely held to a surface; good for biological samples**

- **Disadvantages: more challenging to image in liquids, slower scan speeds needed.**

Non-contact Mode

Non-contact Mode: (attractive Van der Waals) The probe does not contact the sample surface, but oscillates above the adsorbed fluid layer on the surface during scanning. (Note: all samples unless in a controlled UHV or environmental chamber have some liquid adsorbed on the surface). Using a feedback loop to monitor changes in the amplitude due to attractive VdW forces the surfaces topography can be measured.

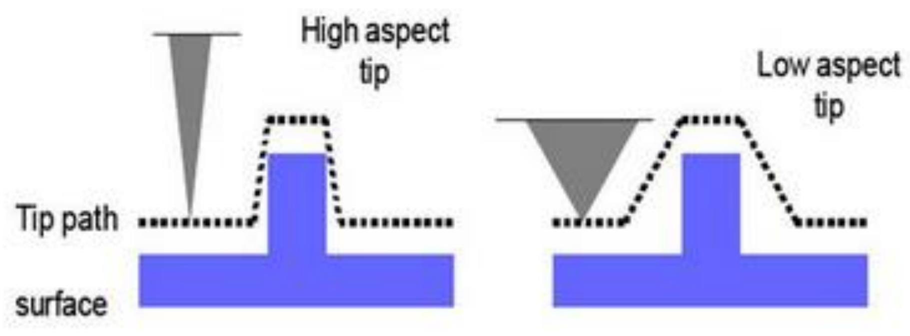


0.5-10 nm probe-surface separation, the forces are attractive

- **Advantages: VERY low force exerted on the samples (10-12N), extended probe lifetime.**
- **Disadvantages: generally lower resolution; contaminant layer on surface can interfere with oscillation; usually need ultra high vacuum (UHV) to have best imaging.**

Limitations of AFM

The AFM can be used to study a wide variety of samples (i.e. plastic, metals, glasses, semiconductors, and biological samples such as the walls of cells and bacteria). Unlike STM or scanning electron microscopy it does not require a conductive sample. **However, there are limitations in achieving atomic resolution. The physical probe used in AFM imaging is not ideally sharp. As a consequence, an AFM imaging does not reflect the true sample topography, but rather represents the interaction of the probe with the sample surface.** This is called tip convolution. **This does not often influence the height of a feature but the lateral resolution.”**

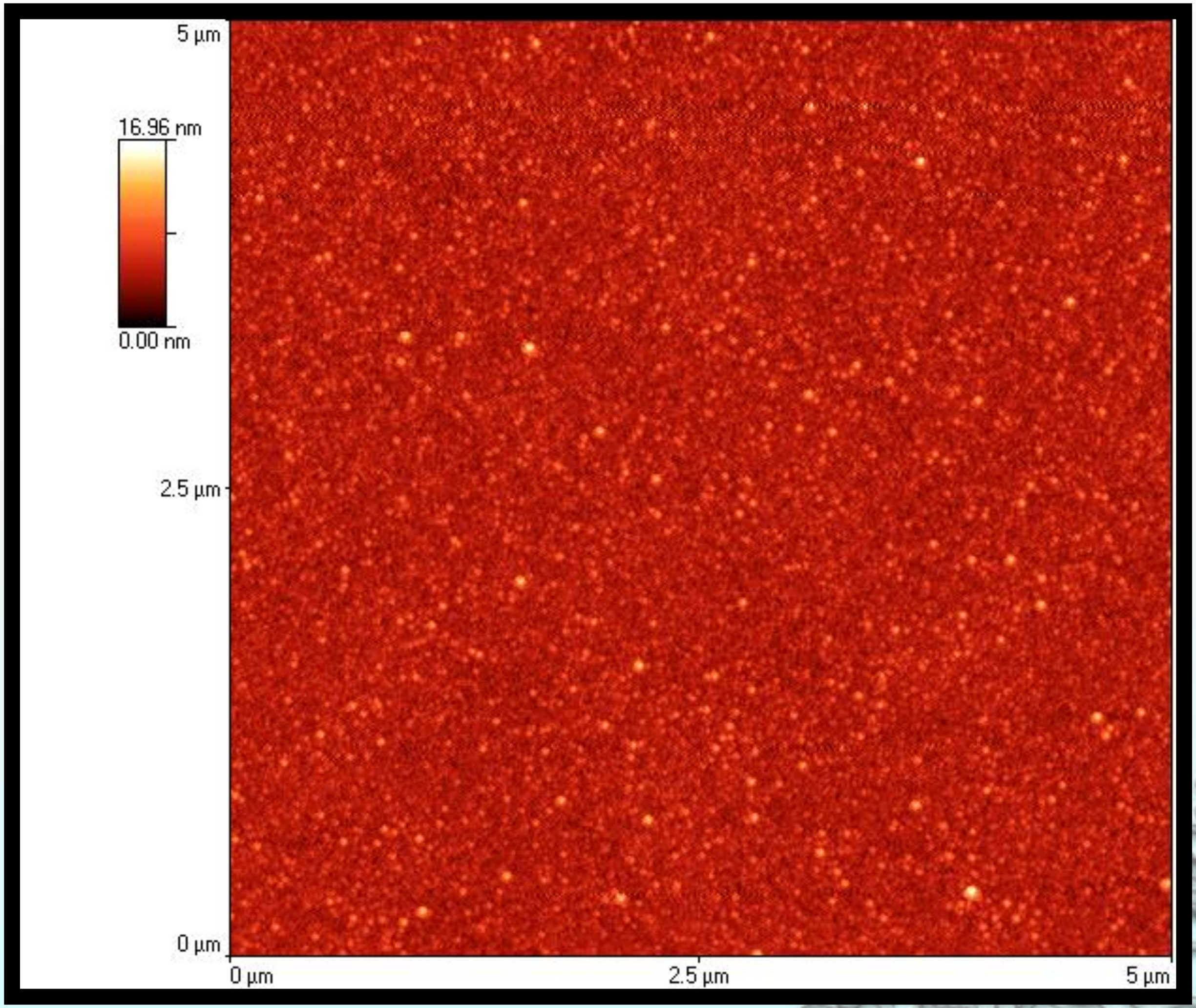


Ideally a probe (tip) with a high aspect ratio will give the best resolution. The radius of curvature of the probe leads to tip convolution. This does not often influence the height of a feature but the lateral resolution.

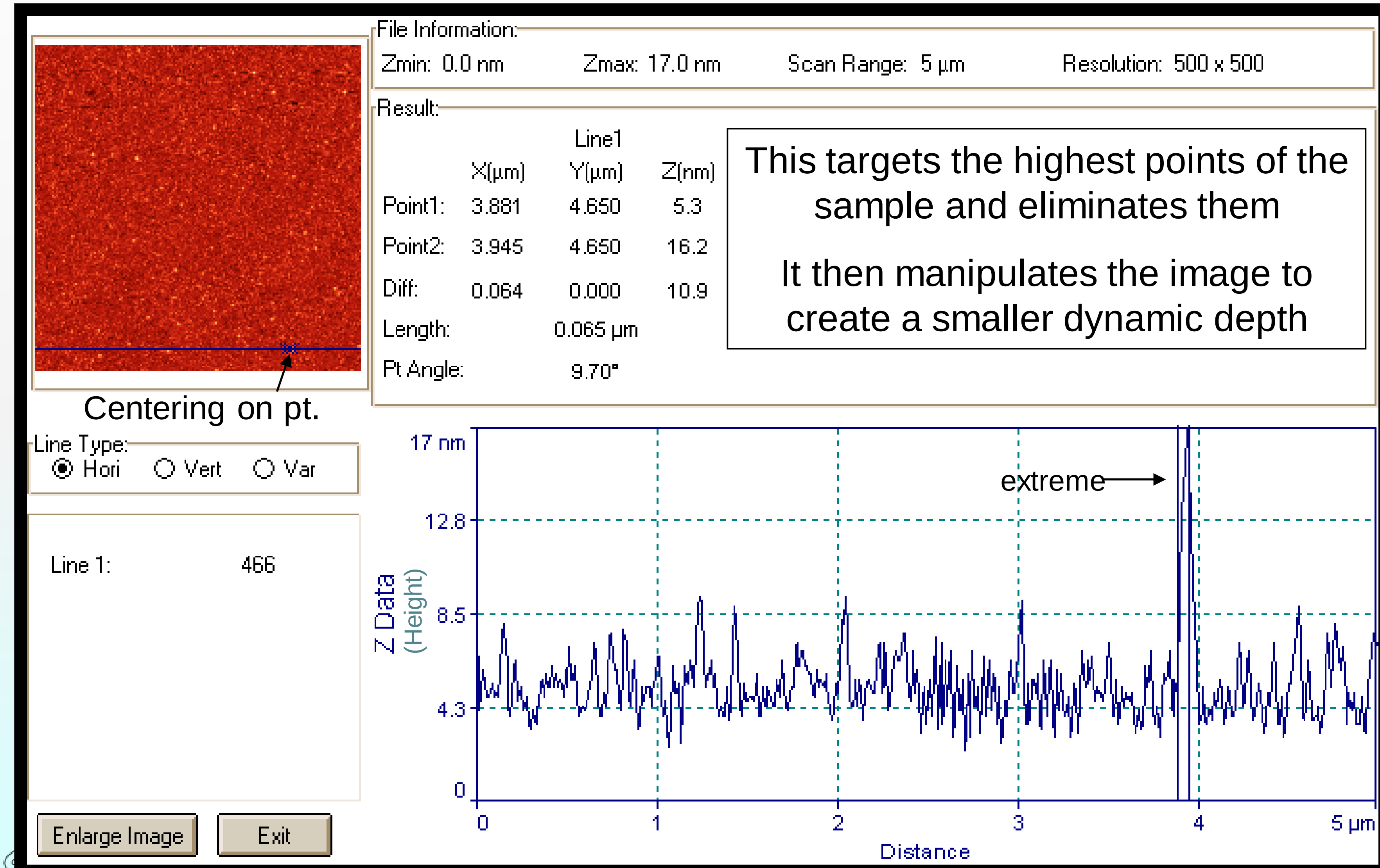
Topography Scanning

Example of generated image upon scanning

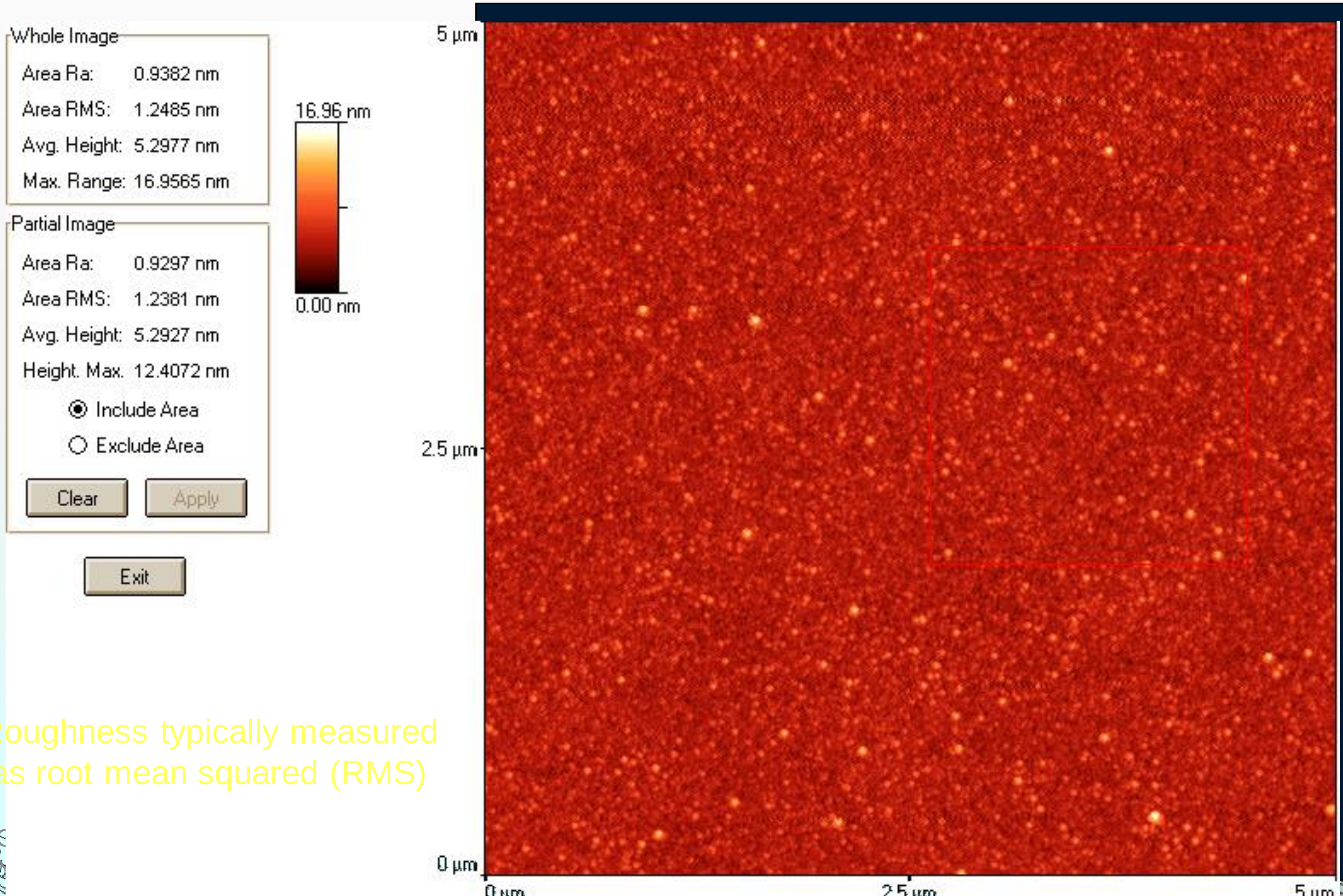
Pd thermally evaporated on Si



Elimination of Extreme Points

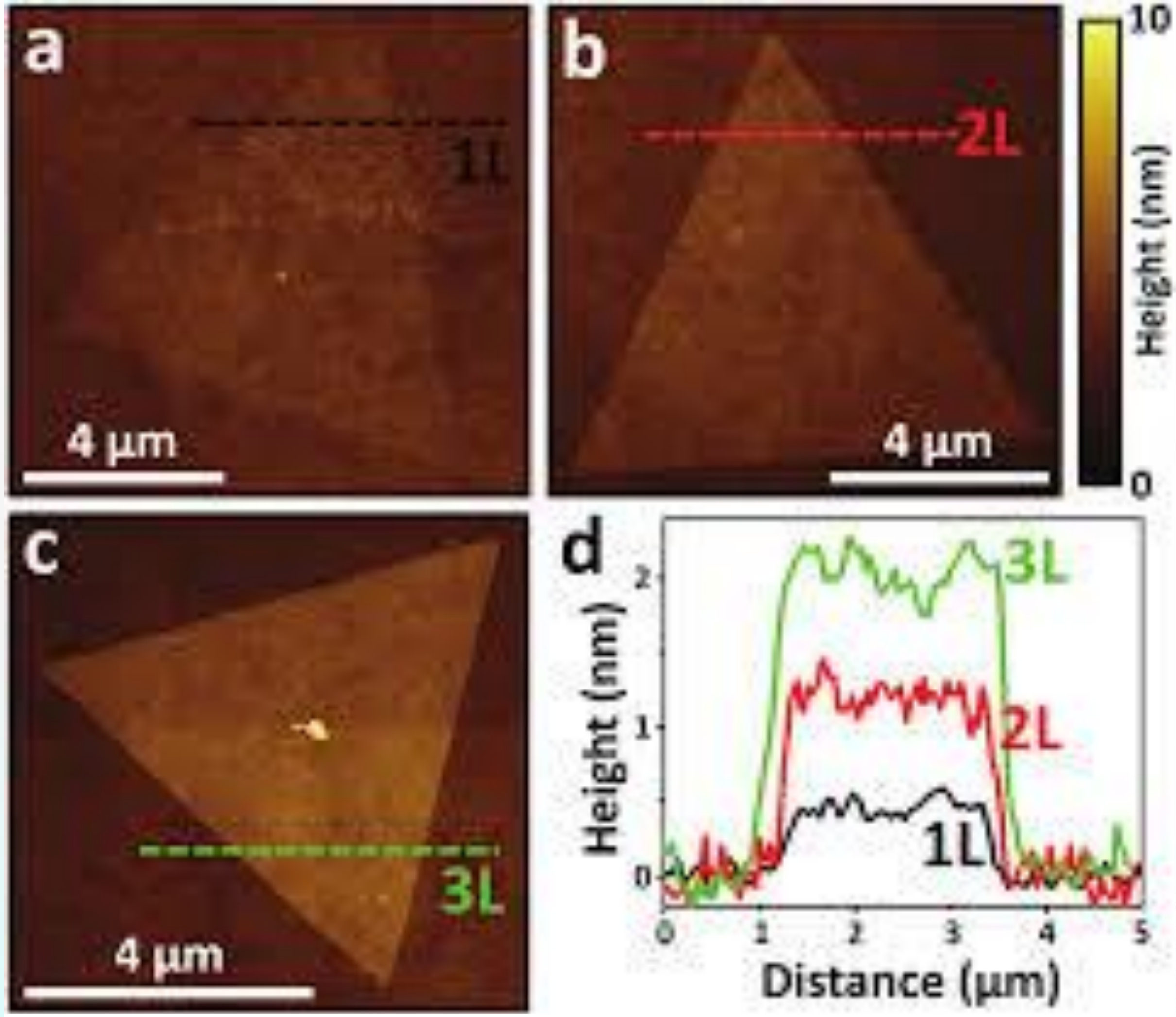


Surface Roughness



Roughness typically measured as root mean squared (RMS)

Thickness measurement



AFM image MoS_2 islands on Si/SiO2. Height profiles measured along the dashed lines.

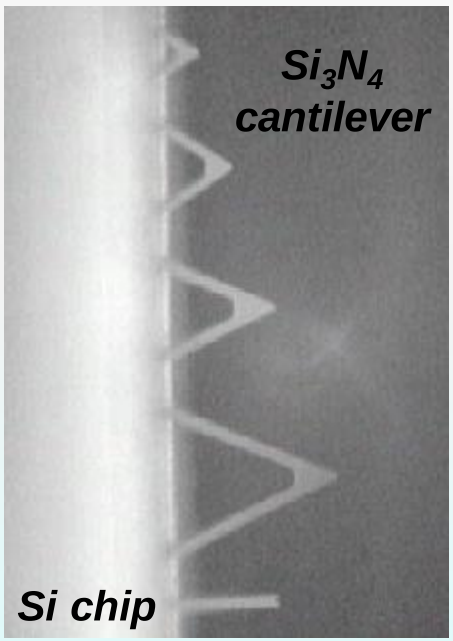
AFM: Tip Functionalization

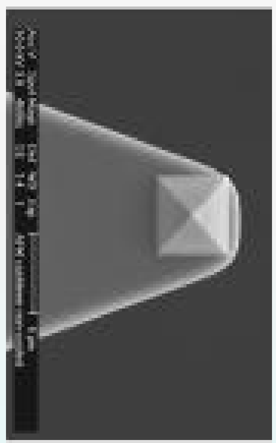
1. Gold coating

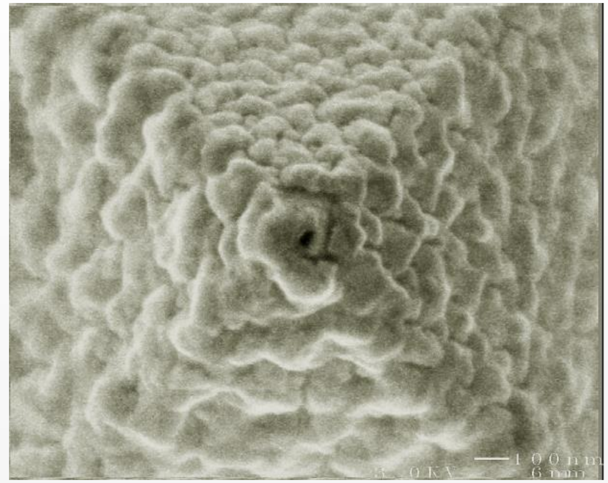
Purpose:

Methods:

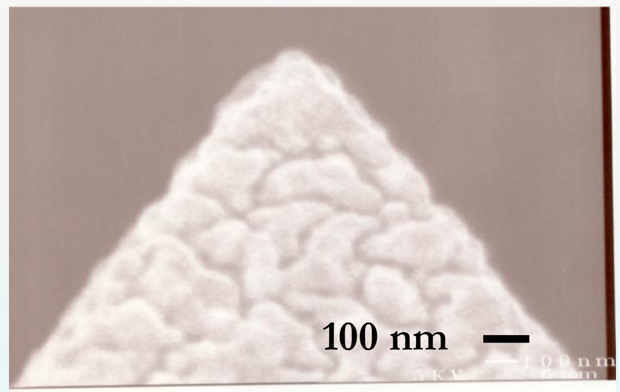
TOP VIEW







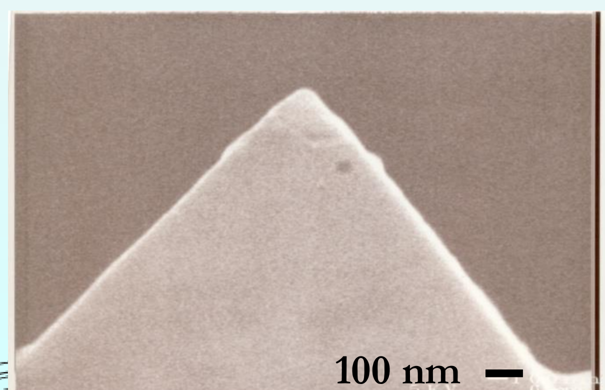
TOP VIEW



ONE-TIME Au COATING :

heterogeneous, rougher larger polydomain microstructure

SIDE VIEW



INTERVAL Au COATING :

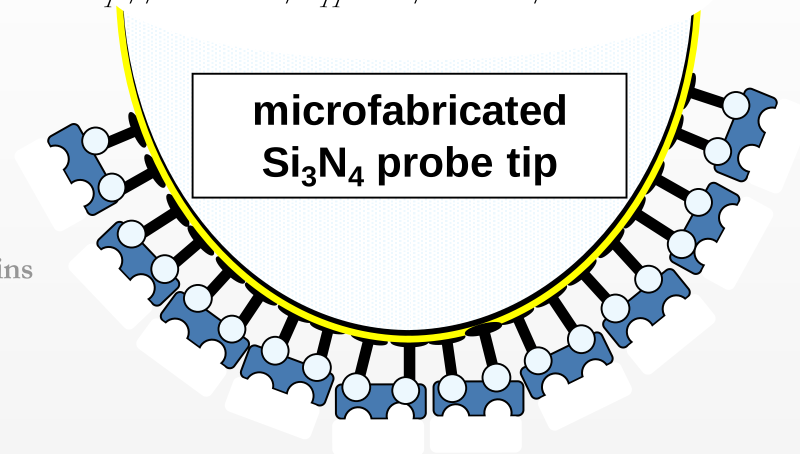
homogeneous, smoother smaller polydomain microstructure

SIDE VIEW

AFM: Tip Functionalization

2. Chemical coating

http://www.di.com/AppNotes/LatChem/LatChemMain.html

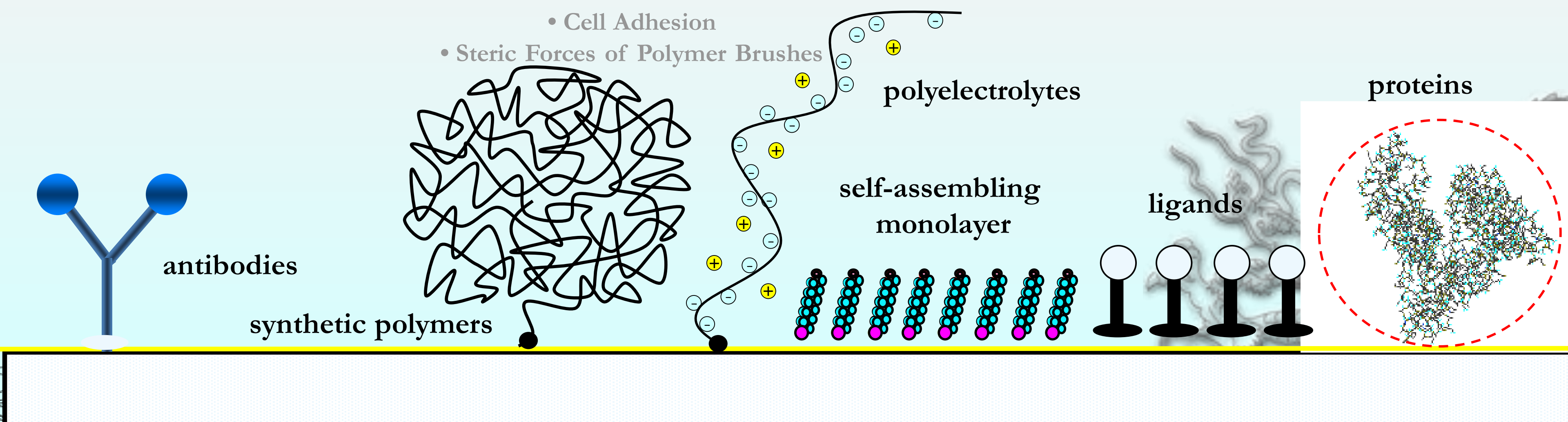


Purpose:

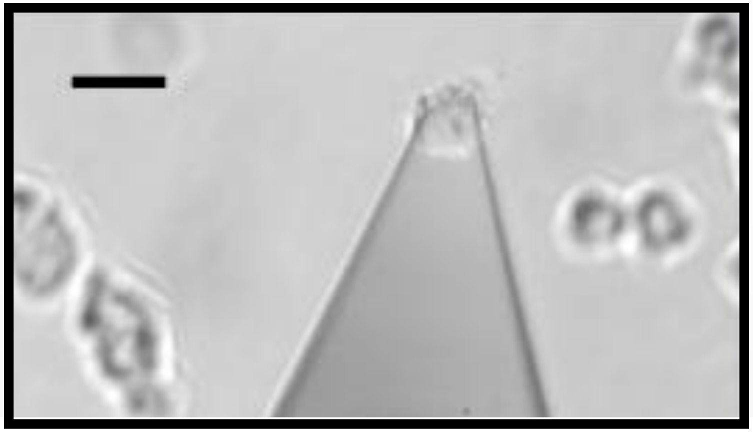
Methods:

Applications:

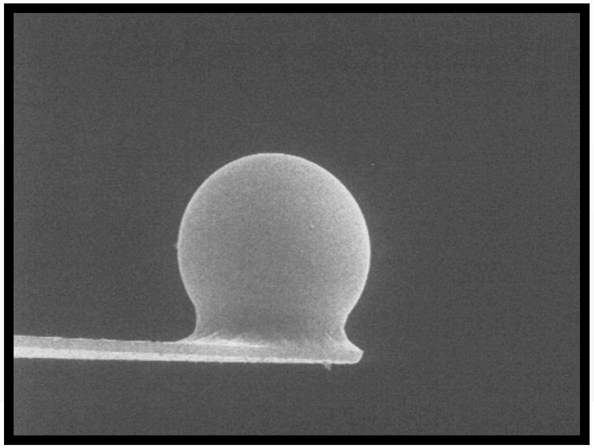
- Molecular Elasticity of Individual Polymer Chains
- Protein Folding
- DNA Interatomic Bonds
- Receptor-Ligand Interactions
- Covalent Bonds
- Colloidal forces
- Van der Waals forces
- Hydration forces
- Hydrophobic forces
- Surface Adhesion
- Nanoindentation
- Electrostatic DLVO forces
- Cell Adhesion
- Steric Forces of Polymer Brushes



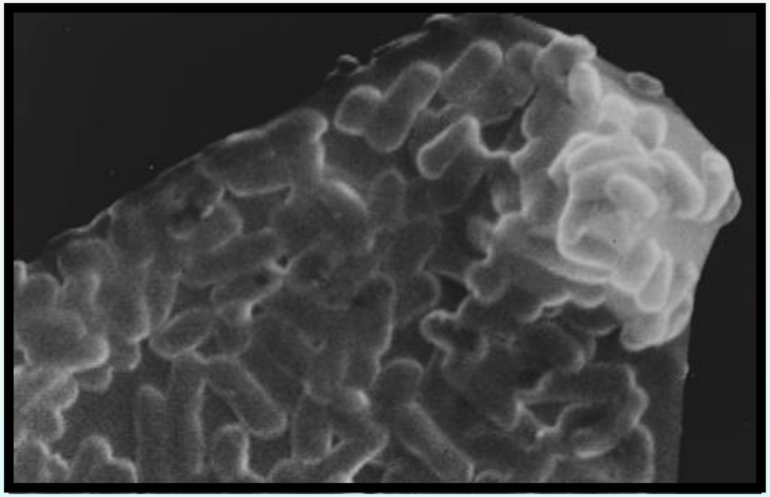
AFM: Tip Functionalization



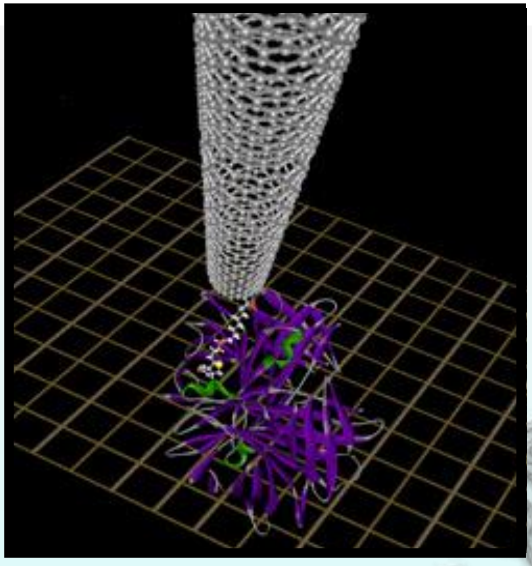
(a) *Single Cell Dictyostelium Discoideum*



(c) *colloidal particle*



(b) *E. Coli Bacteria*



(d) *nanotube with individual ligand*

(a) Benoit, M.; Gabriel, D.; Gerisch, G.; Gaub, H. E. *Nature Cell. Bio* **2000**, *2* (6), 313.
(b) Ong, Y-L.; Razatos, A.; Georgiou, G.; Sharma, M. K. *Langmuir* **1999**, *15*, 2719.
(c) J. Seog, Ortiz/ Grodzinsky Labs 2001
(d) Wong S.S., Joselevich E., Woolley, A.T.; Cheung, C. L., Lieber, C. M. *Nature* **1998**, *394* (6688), 52.

Biological Applications: AFM Images of DNA

TappingMode image of nucleosomal DNA was the highlight of the "Practical Course on Atomic Force Microscopy in Biology," held at the Biozentrum in Basel, Switzerland, July 1998. Image courtesy of Y. Lyubchenko.

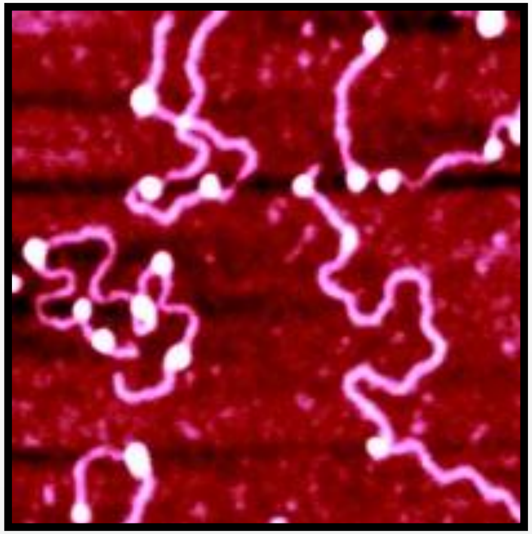
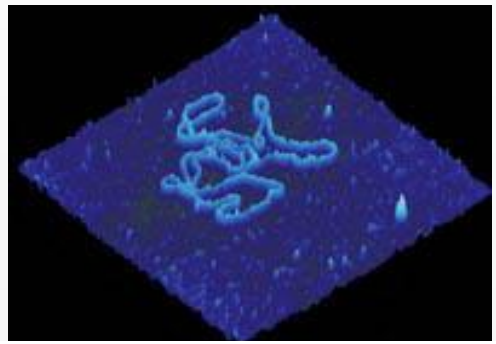
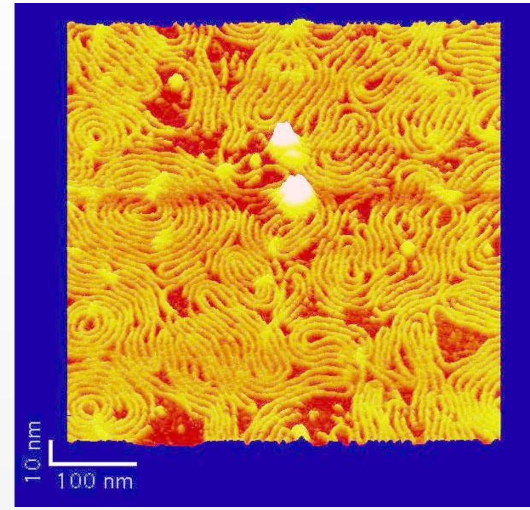


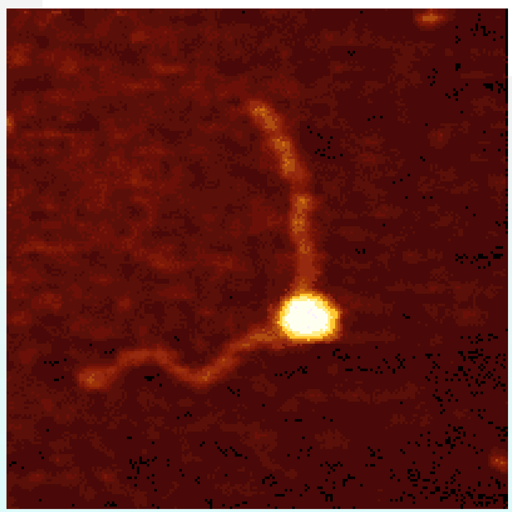
Image of PtyrTlac supercoiled DNA. 750 nm scan courtesy C. Tolksdorf, Digital Instruments/Veeco, Santa Barbara, USA, and R. Schneider and G. Muskhelishvili, Istitut für Genetik und Mikrobiologie, Germany.



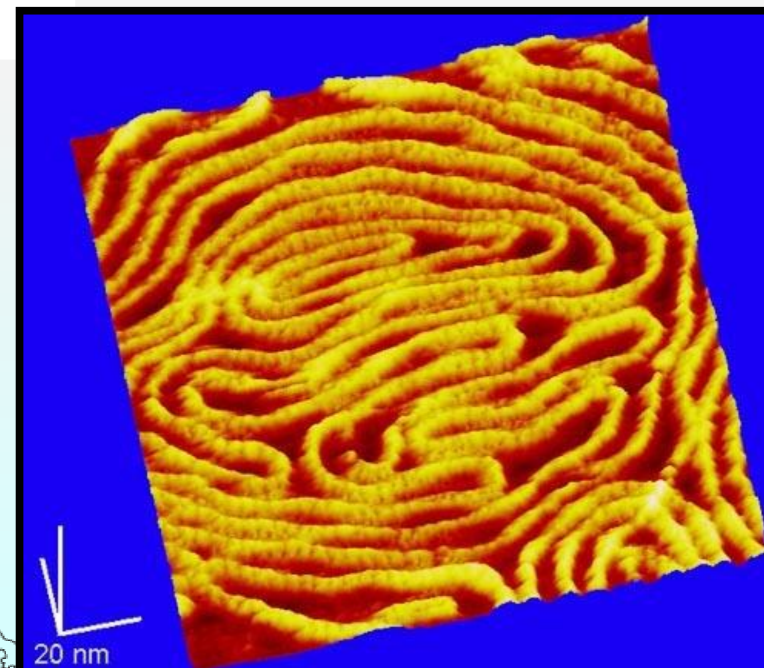


http://www.people.virginia.edu/~js6s/zsfig/DNA.html

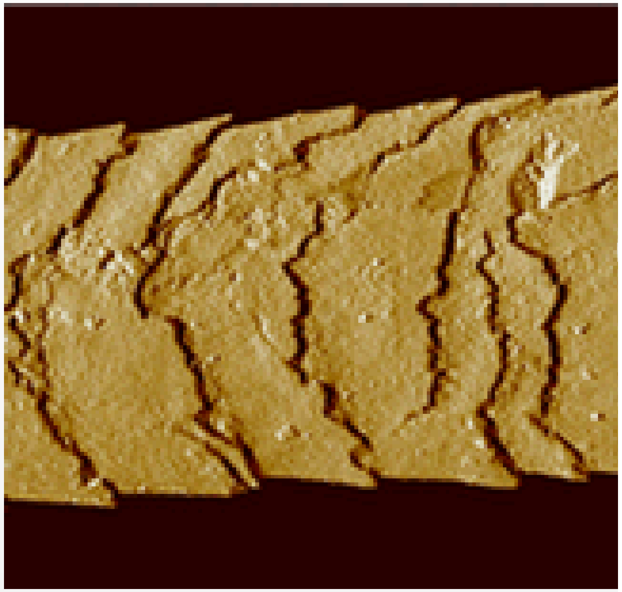
The high resolution of the SPM is able to discern very subtle features such as these two linear dsDNA molecules overlapping each other. 155nm scan. Image courtesy of W. Blaine Stine



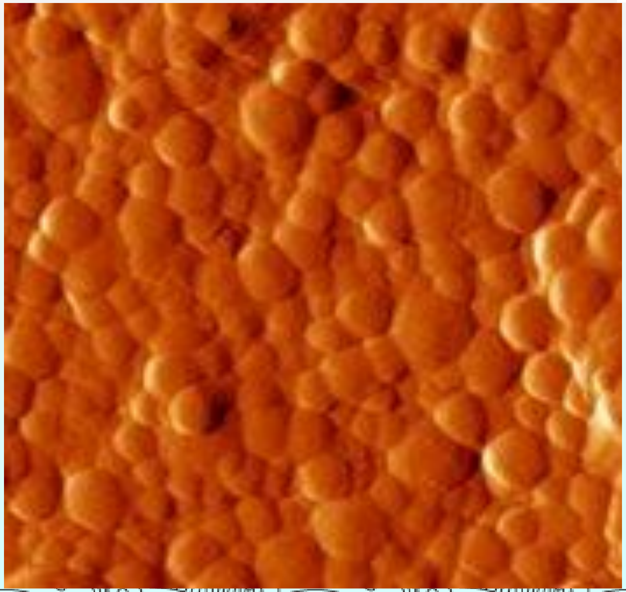
AFM image of short DNA fragment with RNA polymerase molecule bound to transcription recognition site. 238nm scan size. Courtesy of Bustamante Lab, Chemistry Department, University of Oregon, Eugene OR



AFM: From Nano to MicroStructures



Human hair (C. Ortiz)



Eggshell

Other Types of SPM Techniques

- Lateral Force Microscopy (LFM)
  - Frictional forces measured by twisting or “sideways” forces on cantilever.
- Magnetic Force Microscopy (MFM)
  - Magnetic tip detects magnetic fields/measures magnetic properties of the sample.
- Electrostatic Force Microscopy (EFM)
  - Electrically charged Pt tip detects electric fields/measures dielectric and electrostatic properties of the sample
- Chemical Force Microscopy (CFM)
  - Chemically functionalized tip can interact with molecules on the surface – giving info on bond strengths, etc.
- Near Field Scanning Optical Microscopy (NSOM)
  - Optical technique in which a very small aperture is scanned very close to sample
  - Probe is a quartz fiber pulled to a sharp point and coated with aluminum to give a sub-wavelength aperture (~100 nm)

AFM vs SEM

	AFM	SEM
Real 3D image of the studied surface with x, y, z coordinates for each pixel	√	X
Dielectric samples	√	X
Speed	X	√
Samples that can not be exposed to vacuum	√	X
Chemical composition	X	√

Video demo

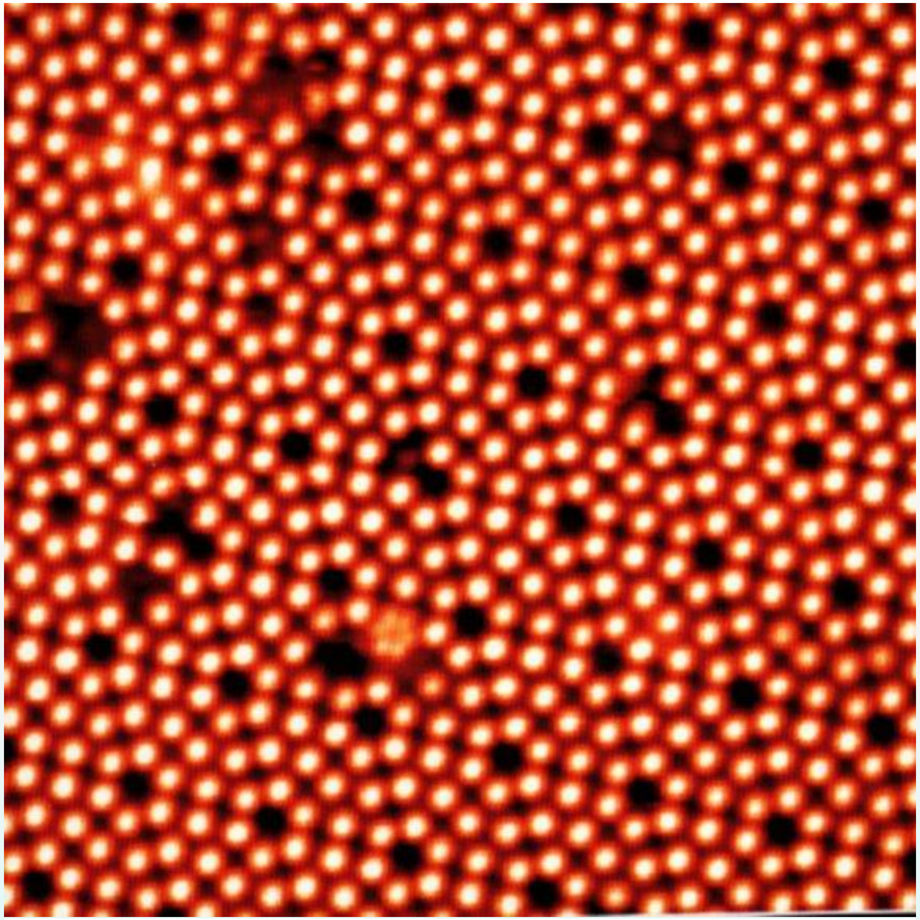
Video 1: how AFM works

Video 2: real lab demo

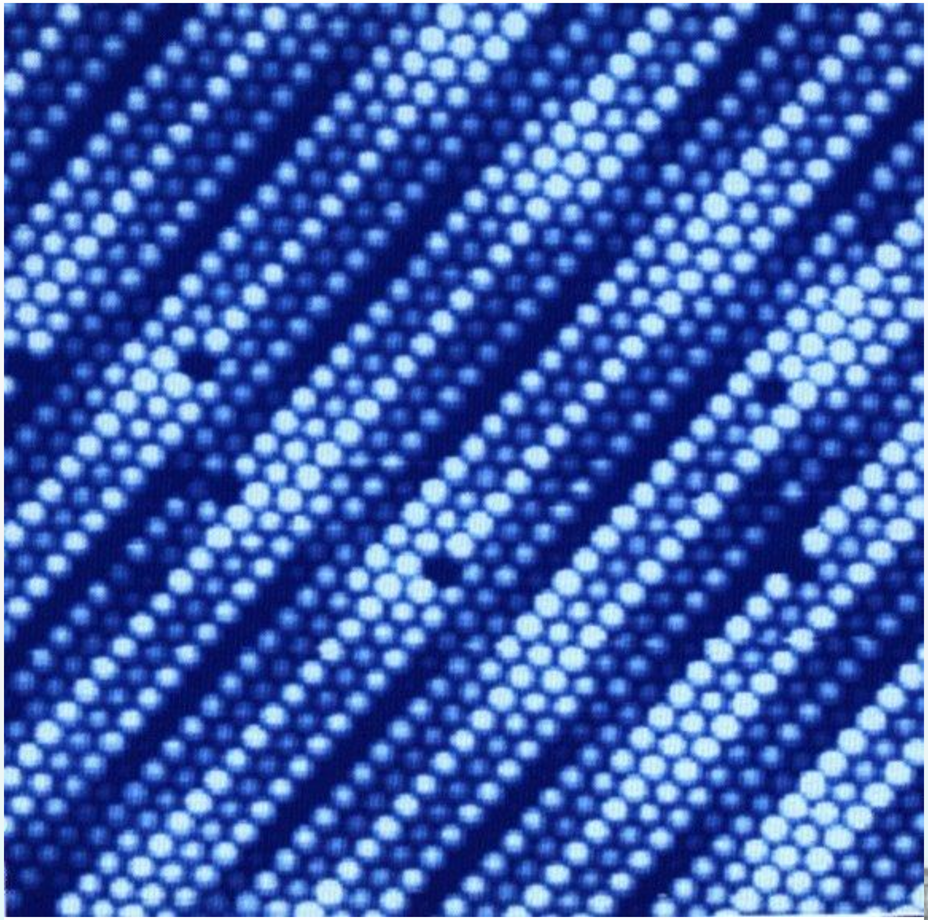
Scanning Tunneling Microscopy

"Seeing" at the nanoscale

STM Images of Point Defects



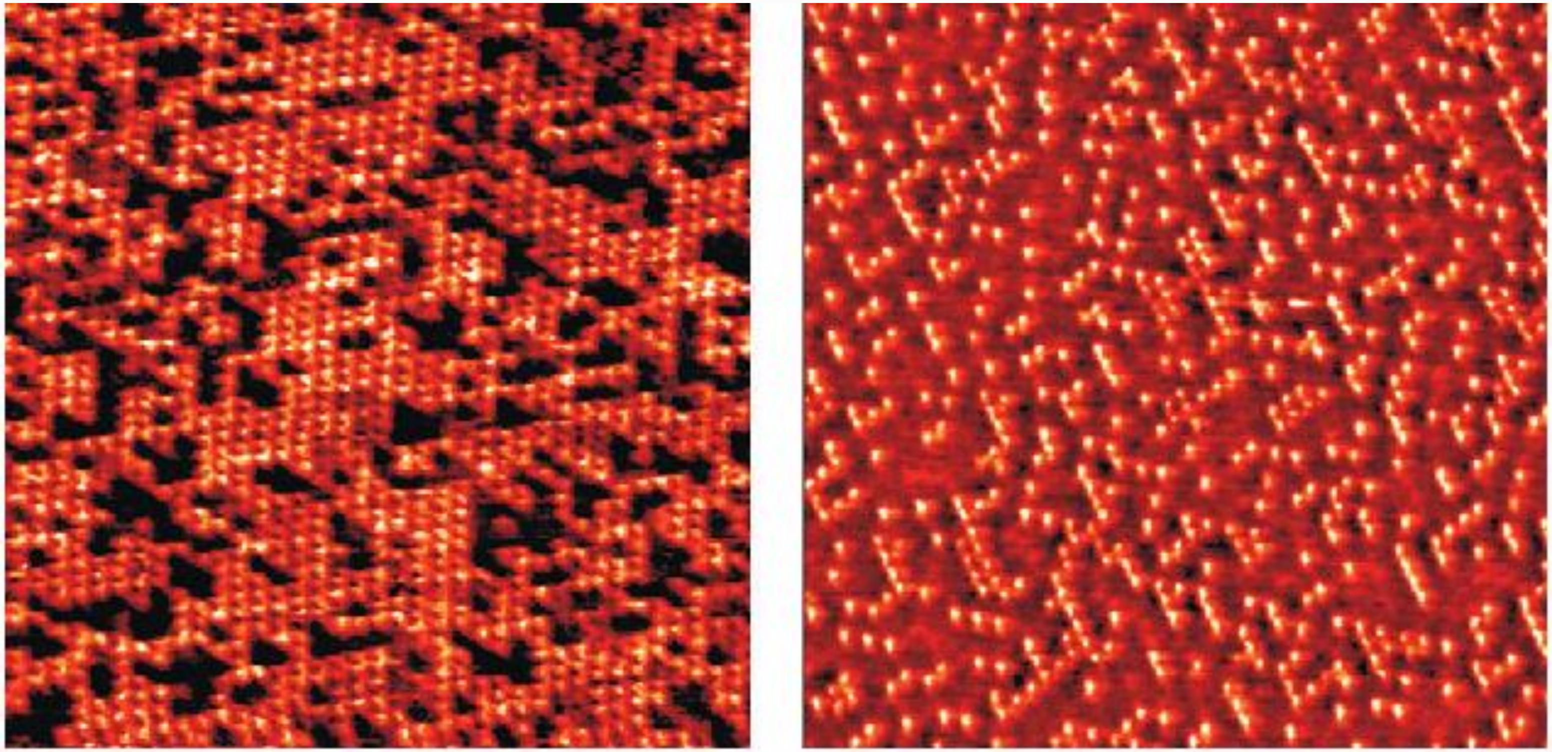
Clean **{111} Si** surface in ultra high vacuum conditions



A STM image of a Pt surface. Vacancies are clearly visible

Source: https://www.tf.unikiel.de/matwis/amat/def_en/kap_4/illustr/g4_2_4.html

STM Images of Alloy



Atomic resolution STM images of a $Pd_{70}Ag_{30}$/Pd(111) (left) and a $Pd_{36}Ag_{64}$/Pd(111) (right) monolayer surface alloy (after annealing to 800 K, image size: 10 nm 10 nm)

Brief History of STM

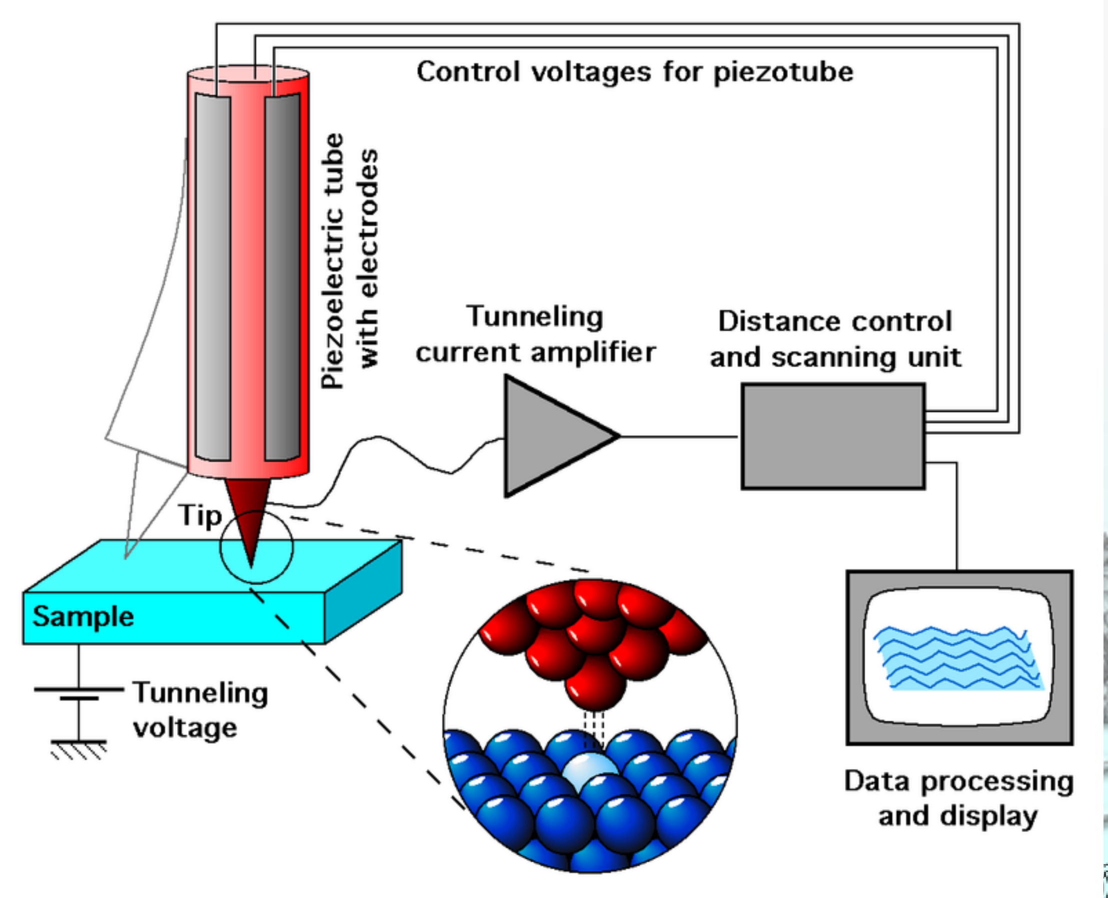
- The first member of SPM family, scanning tunneling microscopy (STM), was developed In 1982, by Gerd Binnig and Heinrich Rohrer at IBM in Zurich created the ideas of STM (Phys. Rev. Lett., 1982, vol 49, p57). Both of the two people won 1986 Nobel prize in physics for their brilliant invention.

STM is really small in size.

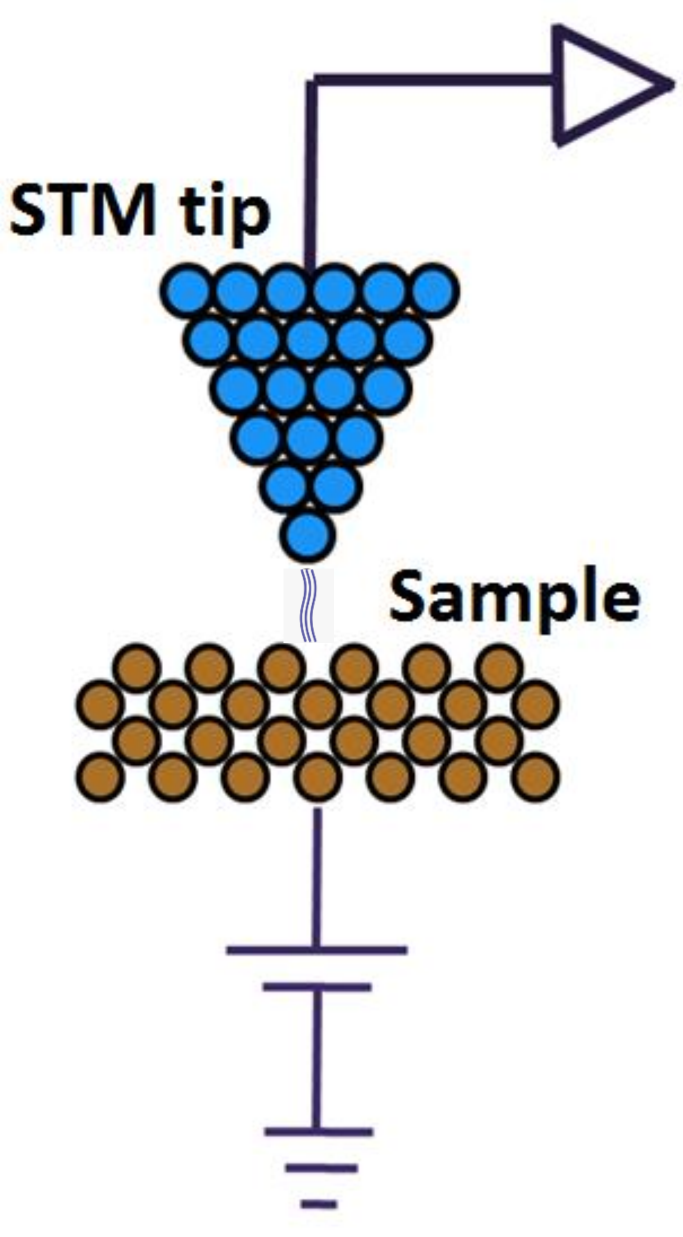
Nobel Laureates Heinrich Rohrer and Gerd Binnig (B. 1947)

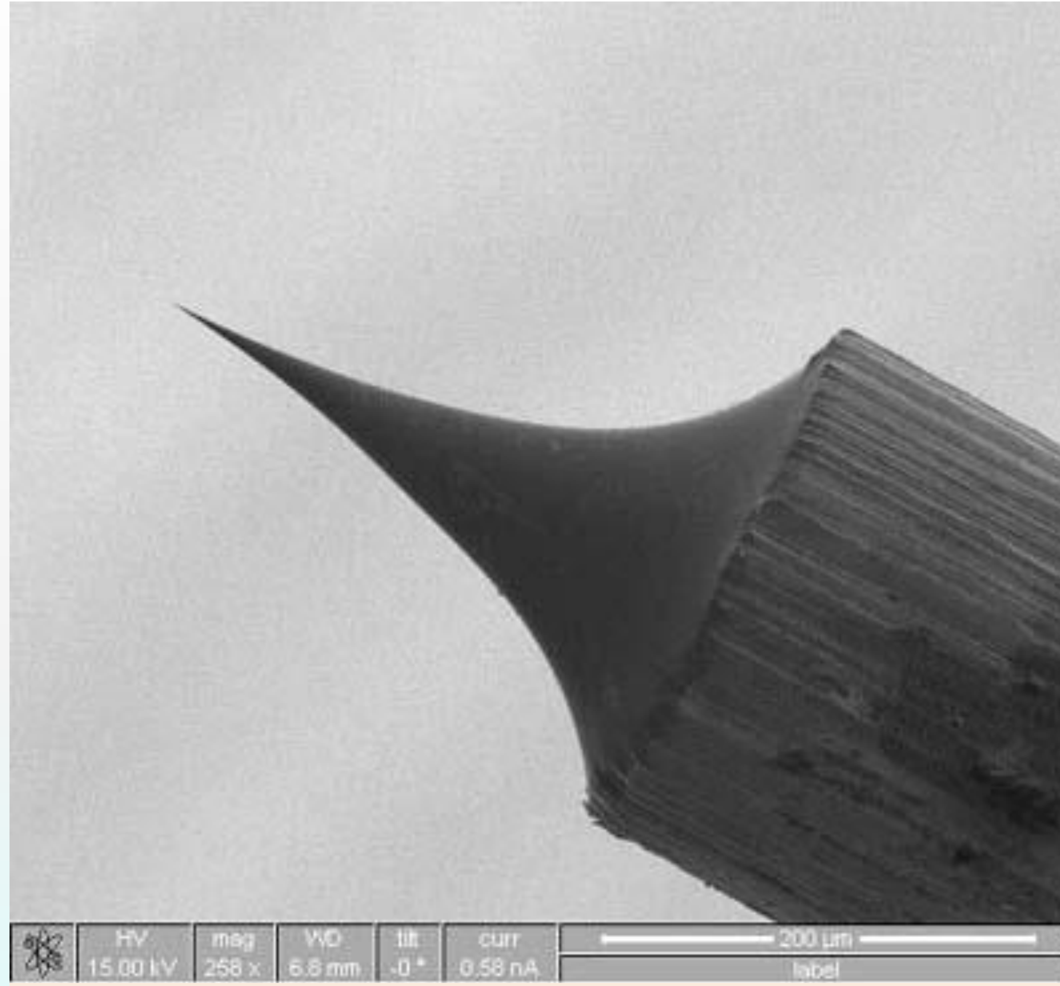
STM

- Similar to AFM, a sharp tip is raster scanned over a sample surface at very short separations (~ 1nm)
- In STM, the signal monitored for the feedback loop is the quantum mechanical tunneling current
- In the most common STM image mode, a piezoelectric scanner moves the tip up and down as it is scanned across the sample to maintain a constant value of this current



STM Tips

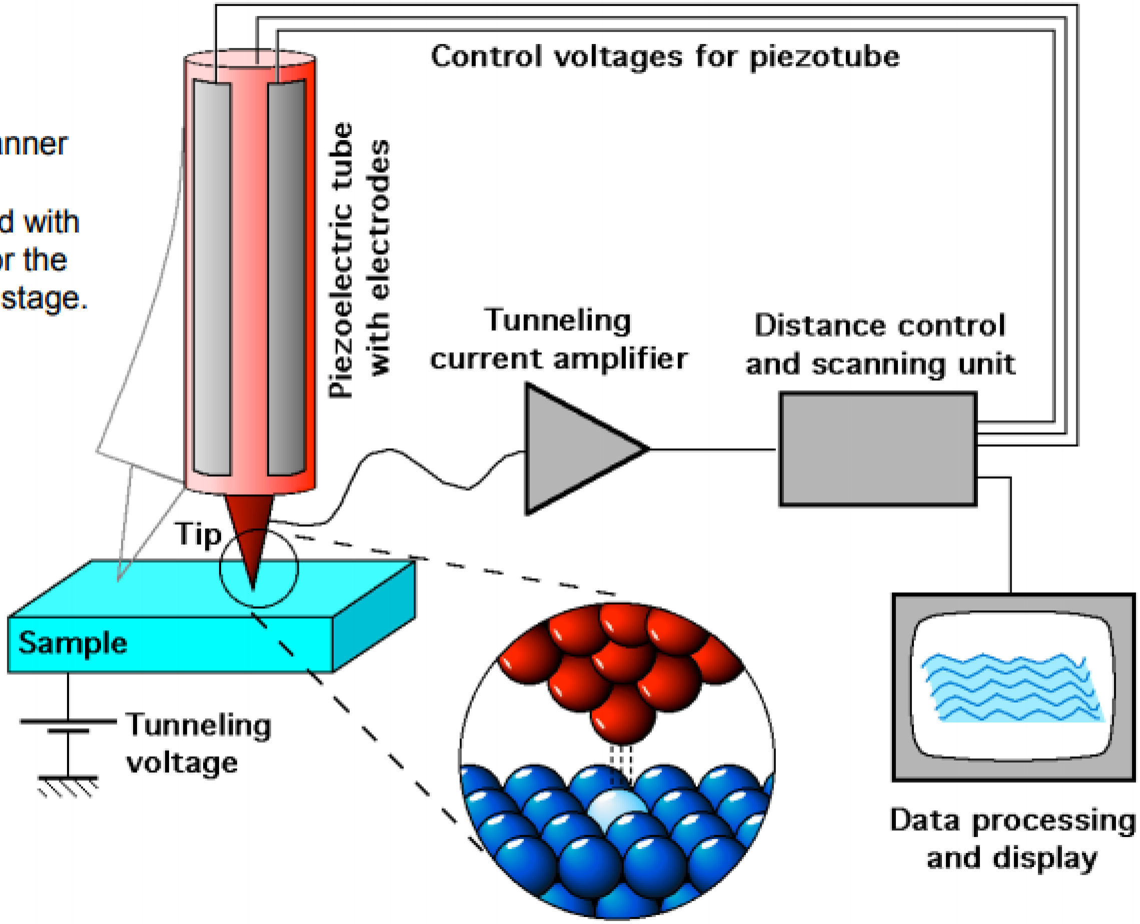




1. Tungsten **tips:** usually made by electrochemical etching
2. Platinum-iridium **tips:** by mechanical shearing.

Basic components of STM:

The scanner can be mounted with the tip or the sample stage.



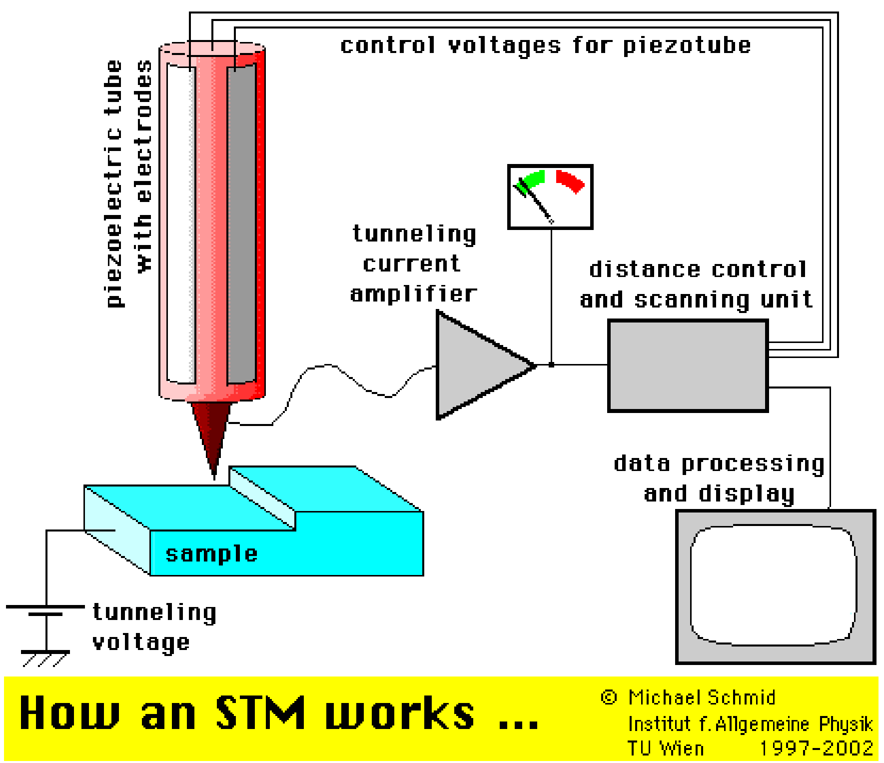
Five basic components:

1. Metal tip,
2. Piezoelectric scanner,
3. Current amplifier (nA),
4. Bipotentiostat (bias),
5. Feedback loop (current).

- Tunneling current from tip to sample or vice-versa depending on bias;
- Current is exponentially dependent on distance;
- Raster scanning gives 2D image;
- Feedback is normally based on constant current, thus measuring the height on surface.

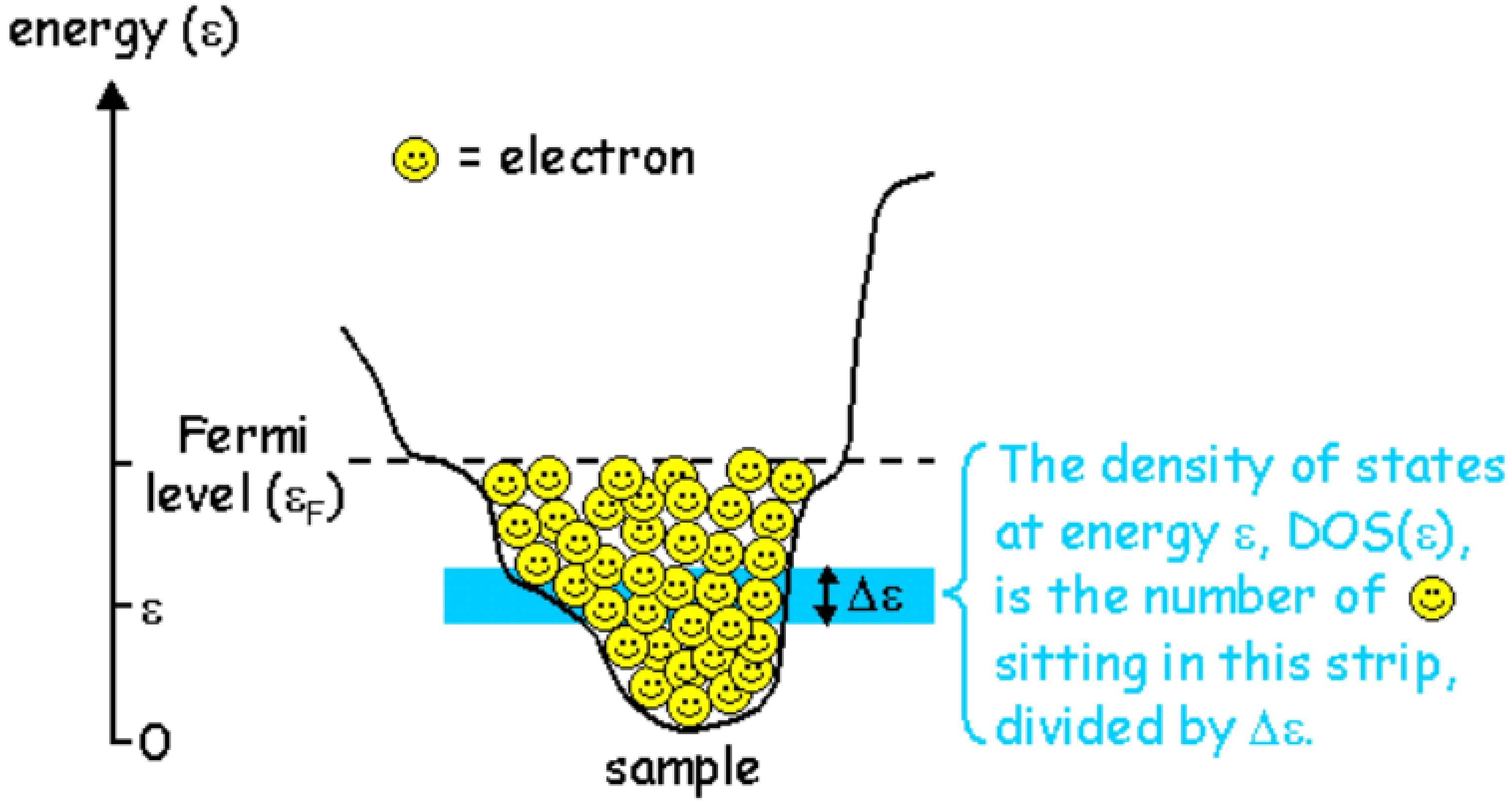
Putting It All Together

- The human hand cannot precisely manipulate at the nanoscale level
- Therefore, specialized materials are used to control the movement of the tip



The **piezoelectric materials** used in **STM** are various kinds of lead zirconate ($PbZrO_3$) or lead titanate ceramics ($PbTiO_3$) since these would have a larger **piezoelectric** coefficient

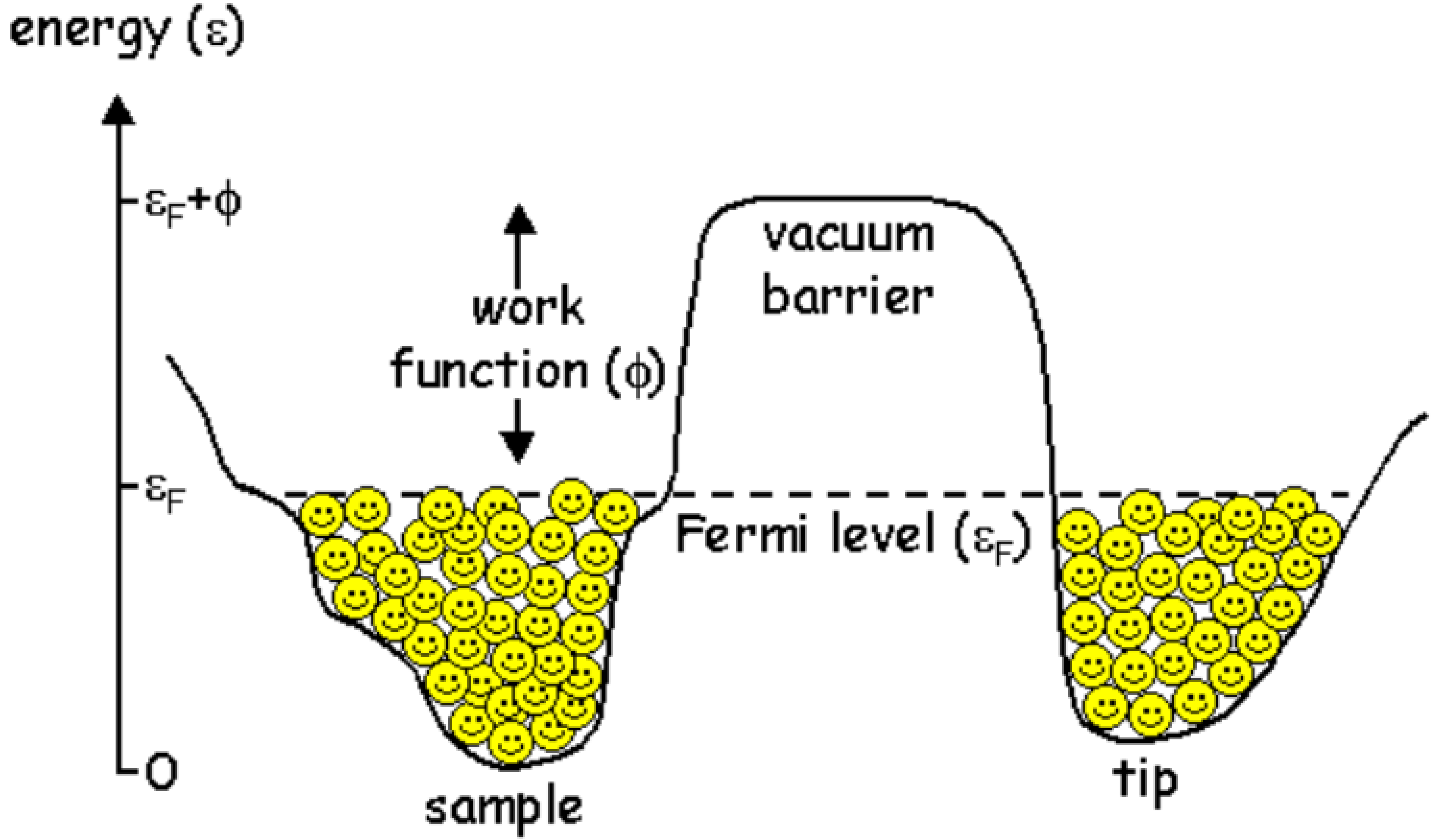
Electron density of states: Fermi level



Electron density of states: Fermi level

- The electrons fill up the energy valley in the sample until there are no more electrons.
- The top energy level at which electrons sit is called the Fermi level, ε_F.
- For every energy ε, the density of states is the number of electrons sitting within $\Delta\varepsilon$ of ε, divided by $\Delta\varepsilon$. So, for the energy shown above as a blue strip, DOS(ε) is approximately $7 / \Delta\varepsilon$.

Tip and Sample: lined up exactly under zero bias

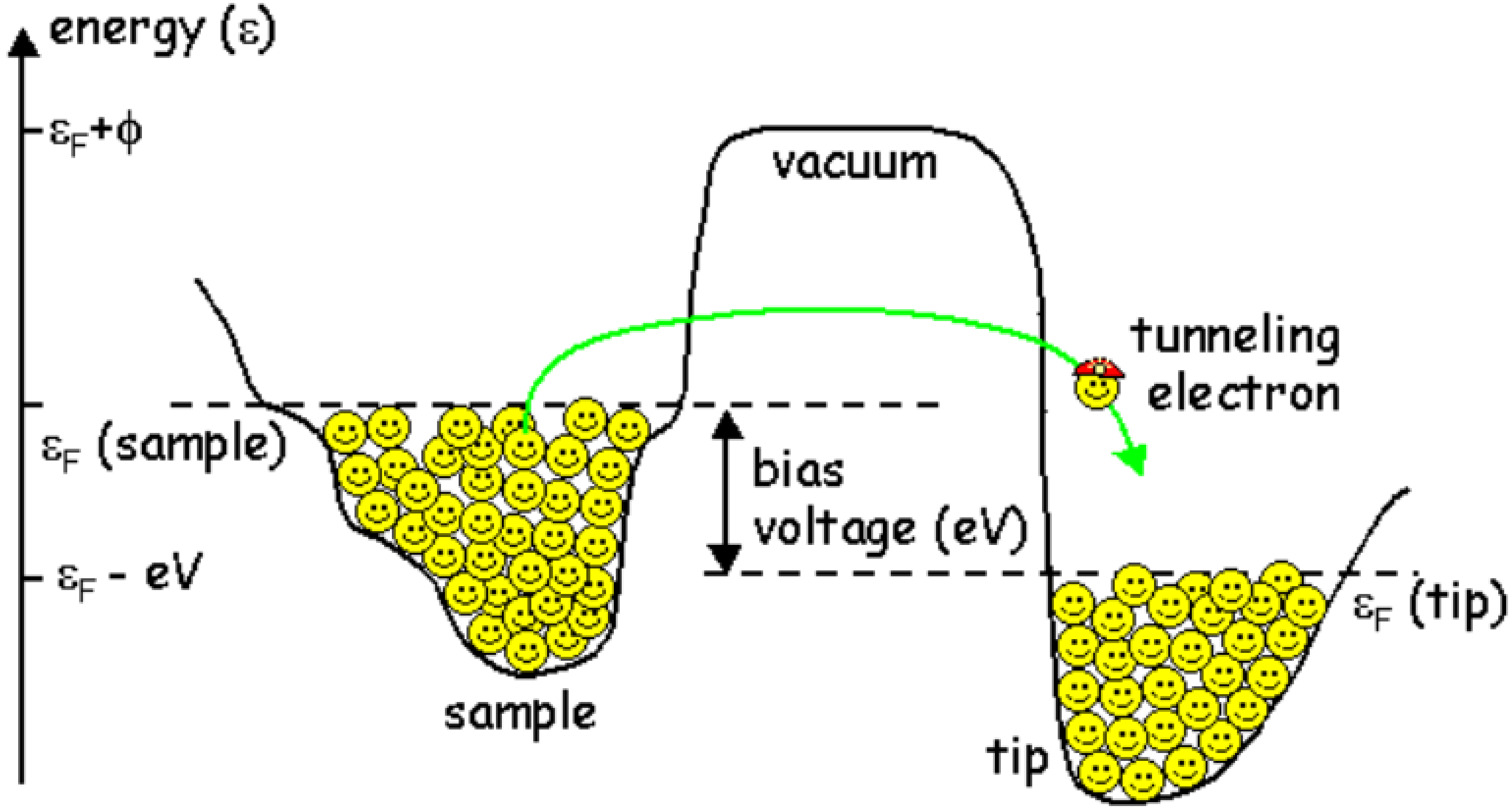


The electrons in the tip and the sample are sitting in two separate valleys, separated by a hill which is the vacuum barrier.

Electron density of states: Fermi level

- Electrons are happy sitting in either the tip or the sample, i.e. they're sitting in nice energy valleys.
- It takes energy to remove an electron into free space. We can think of the vacuum around the tip as an energy hill that the electron would need to climb in order to escape. The height of this energy hill is called the work function, ϕ.
- In order to bring an electron up and *over* the vacuum energy barrier from the tip into the sample (or vice versa), we would need to supply a very large amount of energy.
- Climbing hills is hard work!
- Luckily for us, quantum mechanics tells us that the electron can tunnel right *through* the barrier. Note: *this only works for particles (with both wave and particle characteristics), not for macroscopic objects*. **Don't you try walking through any closed doors! ☺**
- As long as both the tip and the sample are held at the same electrical potential, their Fermi levels line up exactly. There are no empty states on either side available for tunneling into! This is why we apply a bias voltage between the tip and the sample.

Tunneling current at bias

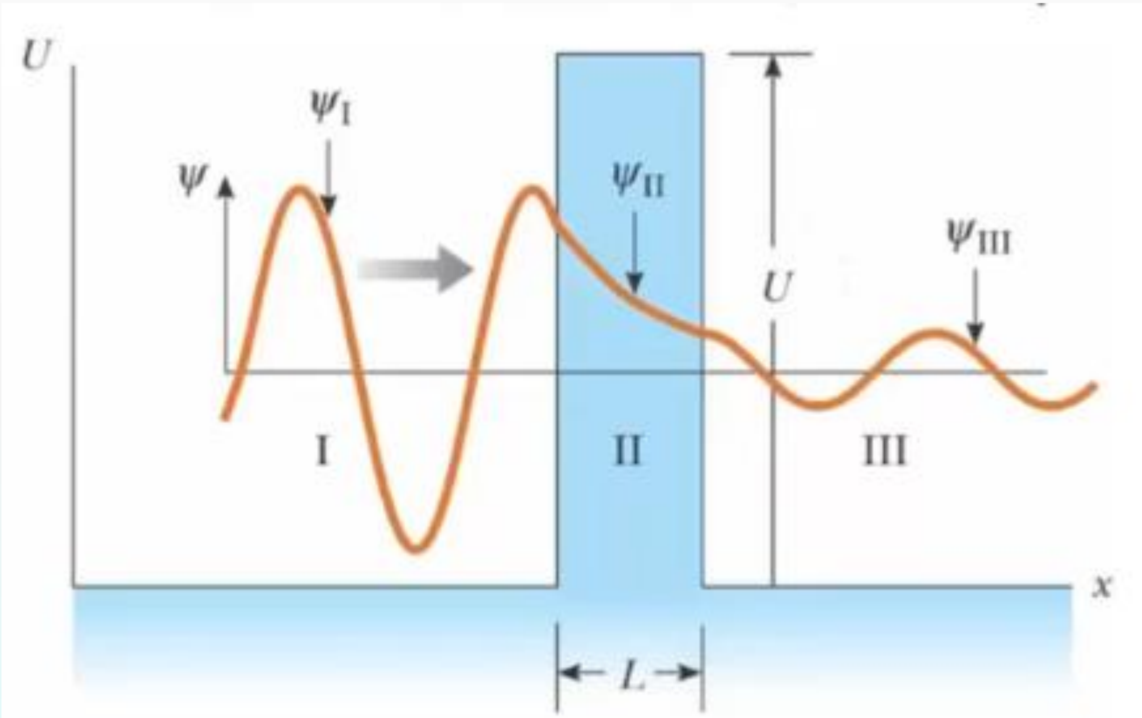


By applying a bias voltage to the sample with respect to the tip, we effectively raise the Fermi level of the sample with respect to the tip. Now we have empty states available for tunneling into.

For a square barrier, $I \approx e^{-2\kappa L}$

$$\kappa = \frac{\sqrt{2m(U-E)}}{\hbar}$$

$\hbar$ is Planck's constant divided by 2π

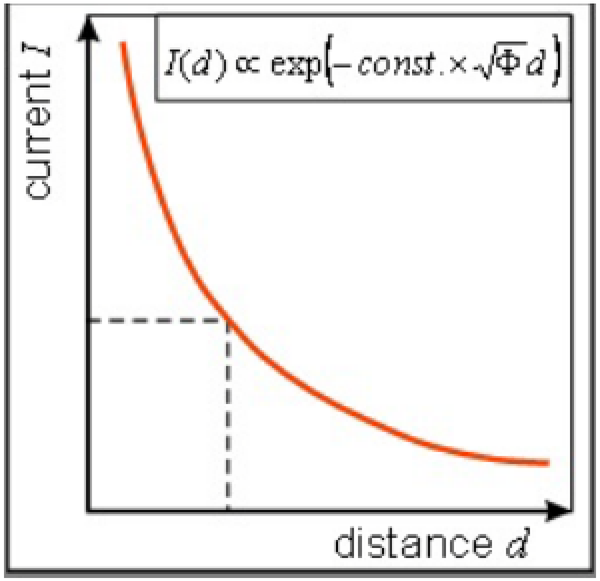


Typical values for U-E are about 4 eV, Currents are 1 nA

Theory of Scanning Tunneling Microscopy: https://arxiv.org/pdf/1404.0961.pdf

Visit for more electron tunneling: https://www2.fkf.mpg.de/ga/research/stmtutor/stmtheo.html

Tunneling Current



$$I(d) = constant \cdot V \cdot e^{(-2\frac{\sqrt{2m\Phi}}{\hbar}d)}$$

Φ: the work function (energy barrier),
e: the electron charge,
m: the electron mass,
h: the Planck's constant,
V: applied voltage,
d: tip-sample distance.

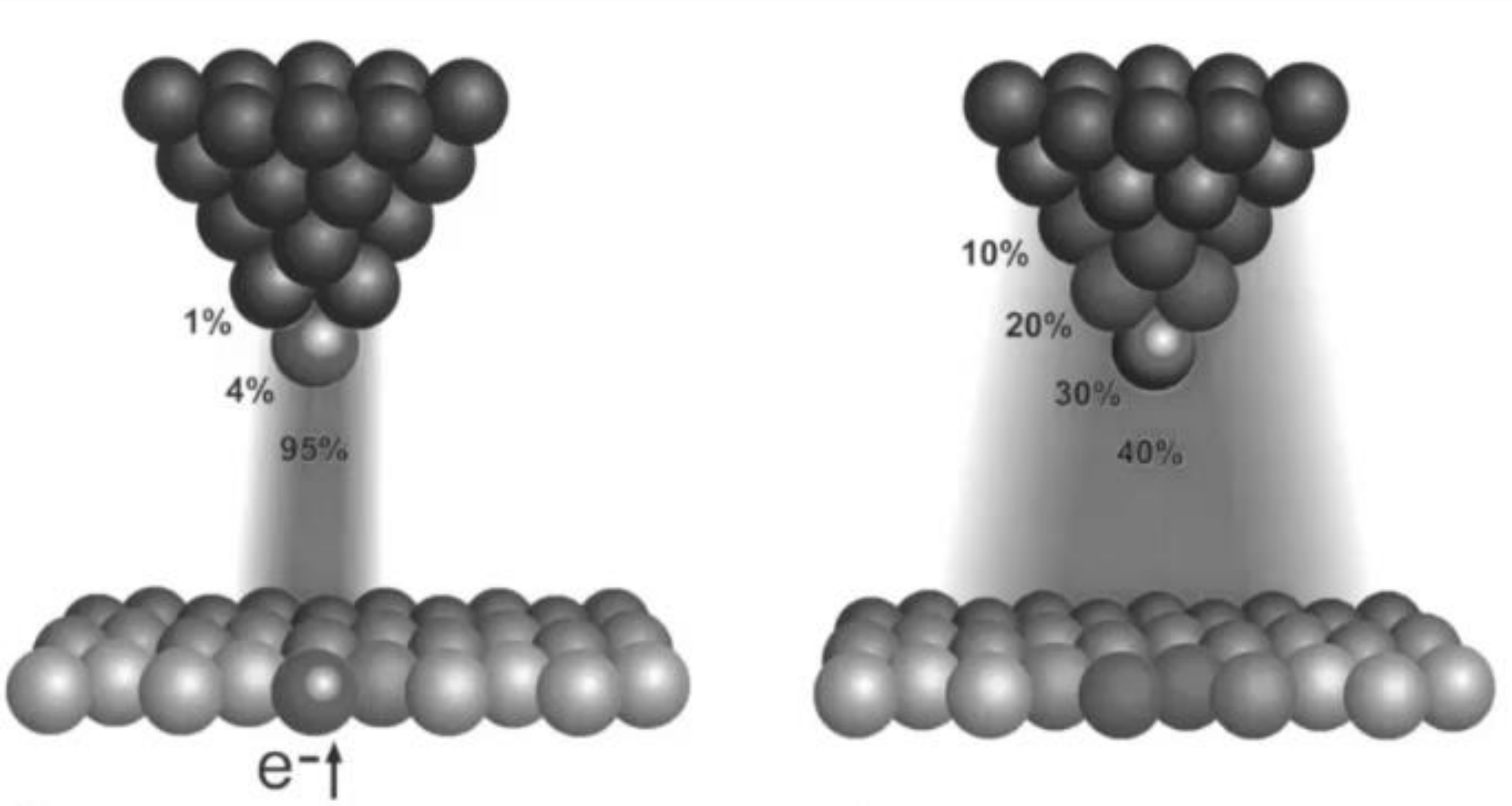
Next page: Φ of common metals.

- A thin metal tip is brought in close proximity of the sample surface. At a distance of only a few Å, the overlap of tip and sample electron wavefunctions is large enough for an electron tunneling to occur.
- When an electrical voltage *V* is applied between sample and tip, this tunneling phenomenon results in a net electrical current, the 'tunneling current'. This current depends on the tip-surface distance *d*, on the voltage *V*, and on the height of the barrier Φ:
- This (approximate) equation shows that the tunneling current obeys Ohm's law, i.e. the current *I* is proportional to the voltage *V*.
- The current depends exponentially on the distance *d*.
- For a typical value of the work function Φ of 4 eV for a metal, the tunneling current reduces by a factor **10** for every **0.1** nm increase in *d*. This means that over a typical atomic diameter of e.g. **0.3 nm**, the tunneling current changes by a factor **1000**! ***This is what makes the STM so sensitive.***
- The tunneling current depends so strongly on the distance that it is dominated by the contribution flowing between the last atom of the tip and the nearest atom in the specimen --- single-atom imaging!

Work Function of Common Metals

Metal	Φ(eV) (Work Function)
Ag (silver)	4.26
Al (aluminum)	4.28
Au (gold)	5.1
Cs (cesium)	2.14
Cu (copper)	4.65
Li (lithium)	2.9
Pb (lead)	4.25
Sn (tin)	4.42
Chromium	4.6
Molybdenum	4.37
Stainless Steel	4.4
Gold	4.8
Tungsten	4.5
Copper	4.5
Nickel	4.6

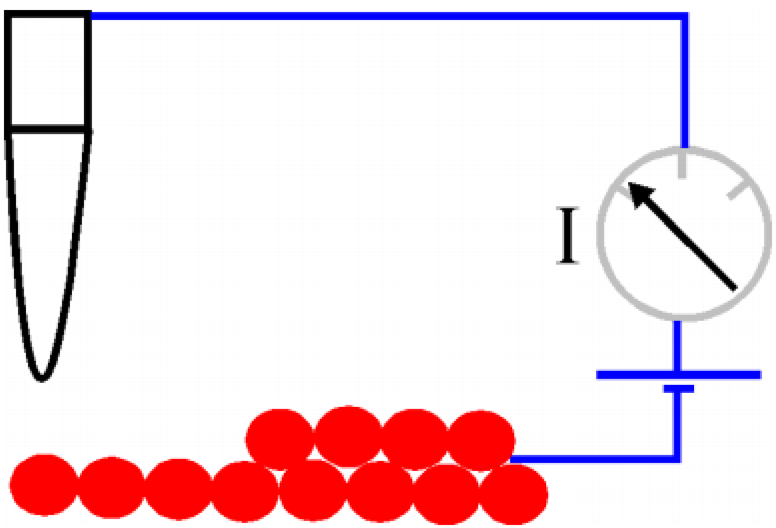
Interatomic action, STM (left) and AFM (right)



An STM gives true atomic resolution. Because the dependence of the tunneling current on the tip-to-sample separation is exponential, only the closest atom on a good STM tip interacts with the closest atom on the sample. For AFM, the dependence of cantilever deflection on tip-to-sample separation is weaker. The result is that several atoms on the tip interact simultaneously with several atoms on the sample

Constant height mode

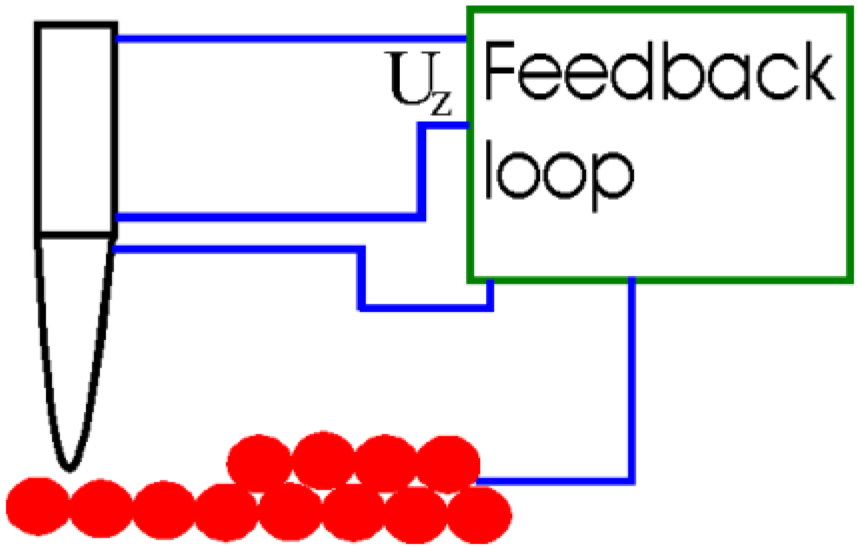
Tell if same atoms



Imaging the different surface atoms (due to their different work functions), revealing the surface composition or defects.

Constant current mode

Tell heights for the same atoms

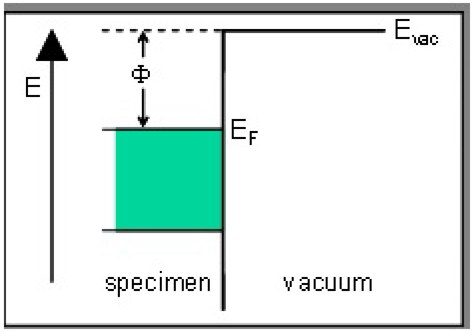


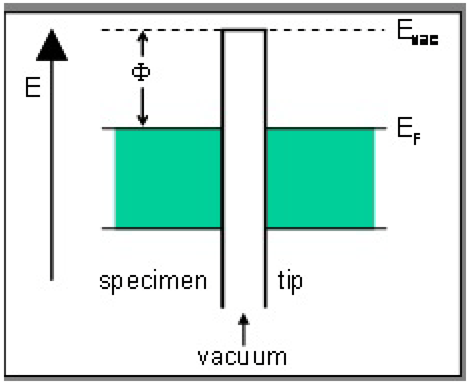
Imaging the surface topography at atomic resolution if the surface is composed of the same atoms, *i.e., the only factor affecting the tunneling current is the distance.*

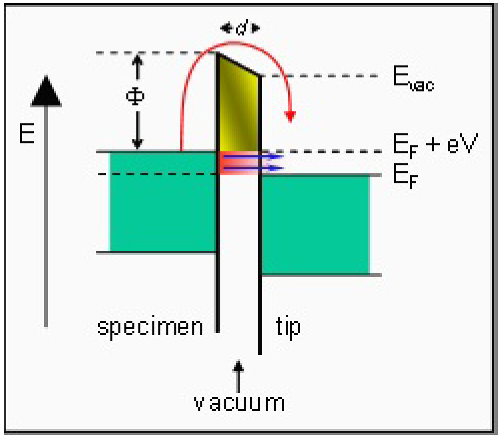
Tunneling Current

a result of the overlap of tip and sample electron wavefunctions

Two requirements: 1. small distance --- electron wavefunction overlap
2. bias --- for net current flow.



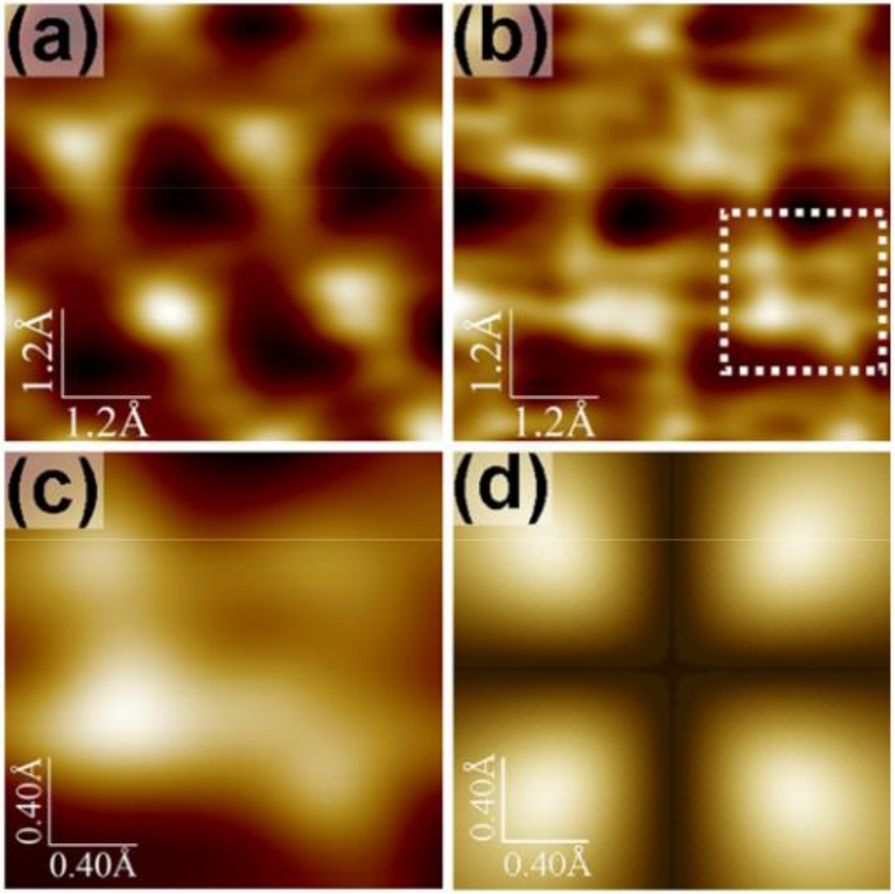




In a metal, the energy levels of the electrons are filled up to a particular energy, known as the 'Fermi energy' E_F. In order for an electron to leave the metal, it needs an additional amount of energy Φ, the so-called 'work function'.

When the specimen and the tip are brought close to each other, there is only a narrow region of empty space left between them. On either side, the electrons are present up to the Fermi energy. They need to overcome a barrier Φ to travel from tip to specimen or vice versa.

If the distance d between specimen and tip is small enough, electrons can 'tunnel' through the vacuum barrier. When a voltage V is applied between specimen and tip, the tunneling effect results in a net electron current. In this example from specimen to tip. This is the tunneling current.



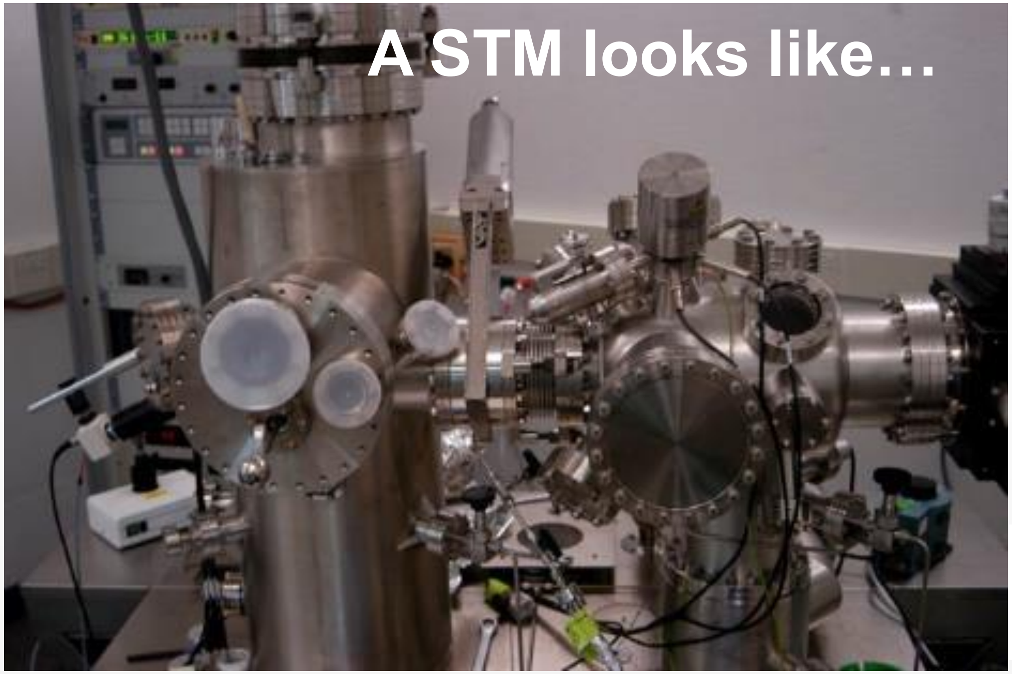
Selecting the tip electron orbital for scanning tunneling microscopy imaging with sub-ångström lateral resolution

EPL,92(2010) 46003

Fig. 5: (Color online) (a), (b) $6 \times 6\,\text{Å}^2$ STM images of HOPG(0001) surface measured with W[001] tip at $I = 0.7\,\text{nA}$ (a) and $I = 1.6\,\text{nA}$ (b). $U = -100\,\text{mV}$. (c) A $2 \times 2\,\text{Å}^2$ region indicated by the dashed square in panel (b). (d) Electron probability density distribution for the W $5d_{x^2-y^2}$-orbital in the plane $z = 1.0\,\text{Å}$ from the nucleus. This plane was selected because of necessity of extremely small tip-sample separations for subatomic imaging and due to the fact that carbon orbitals are less localized than tungsten orbitals (see text and fig. 8).

What do you have to do to get images like that?

- Mechanical vibration isolation
- Noise isolation
- Electrically isolated
- Thermally stable (and often very cold, liquid helium temperatures)
- Ultra high vacuum, to avoid contaminants



[Nanolino - Departement of Physics – Univ. Basel]

Want to learn more about STM?

http://hoffman.physics.harvard.edu/ https://capricorn.bc.edu/wp/zeljkoviclab/

STM Tips

- STM tip should be conducting (metals, like Pt);
- STM plays with the very top (outermost) atom at the tip and the nearest atom on sample; so the whole tip is not necessarily very sharp in shape, different from the case of AFM, where spatial "contact" is necessary and crucial for feedback.
- **How do we obtain these wonderful tunneling tips where only one atom is at the top?**

  Answer: really easy to obtain such tips, simply by cutting a thin metal wire using a wire cutter --- *there is always a single atom left over at the very top.*

Challenges of the STM

- Works primarily with conducting materials
- Vibrational interference
- Contamination
  - Physical (dust and other pollutants in the air)
  - Chemical (chemical reactivity)

Vertical resolution:

- 1-10 pm (10^{-12} m)

Lateral resolution:

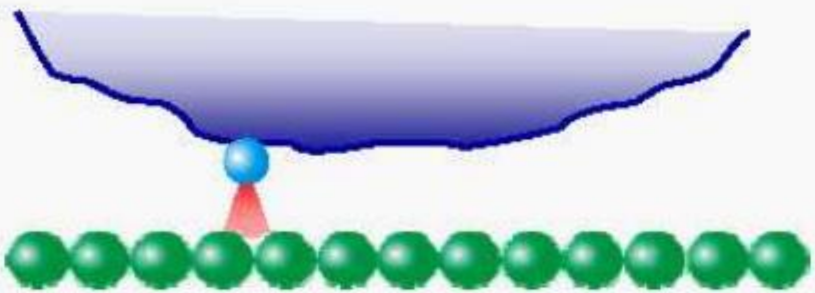
- 100 pm (10^{-10} m)

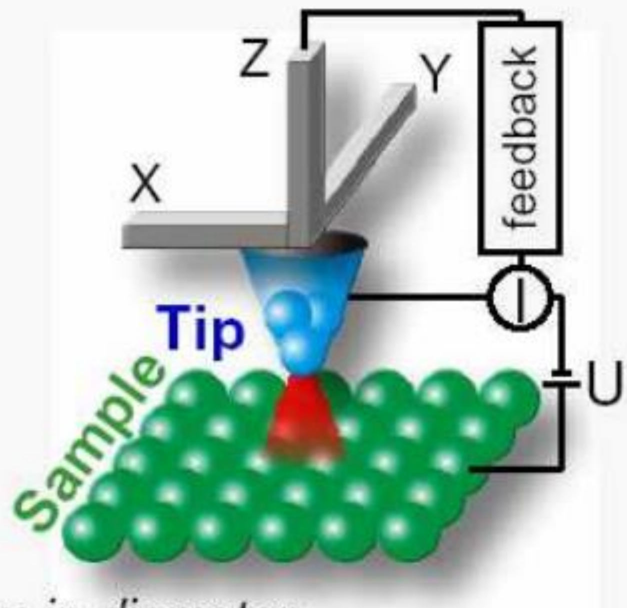
Resolution limitations:

- vibrations
- temperature
- tip stability
- surface condition

STM tips:

- metallic wire
- typical wire diameter: 0.2-0.5mm
- mechanically or chemically etched tip
- typical apex diameter: few nm



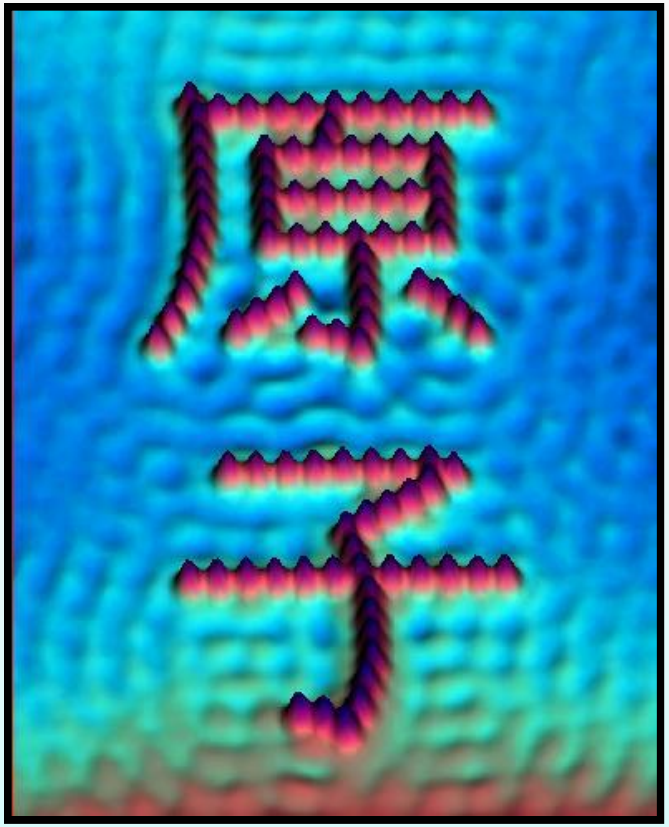


- *~90% of the current flows in a channel 0.1nm in diameter*
  → single atom at apex of broad tip still enables atomic-scale resolution
- but, poor resolution if atom is unstable and jumps around

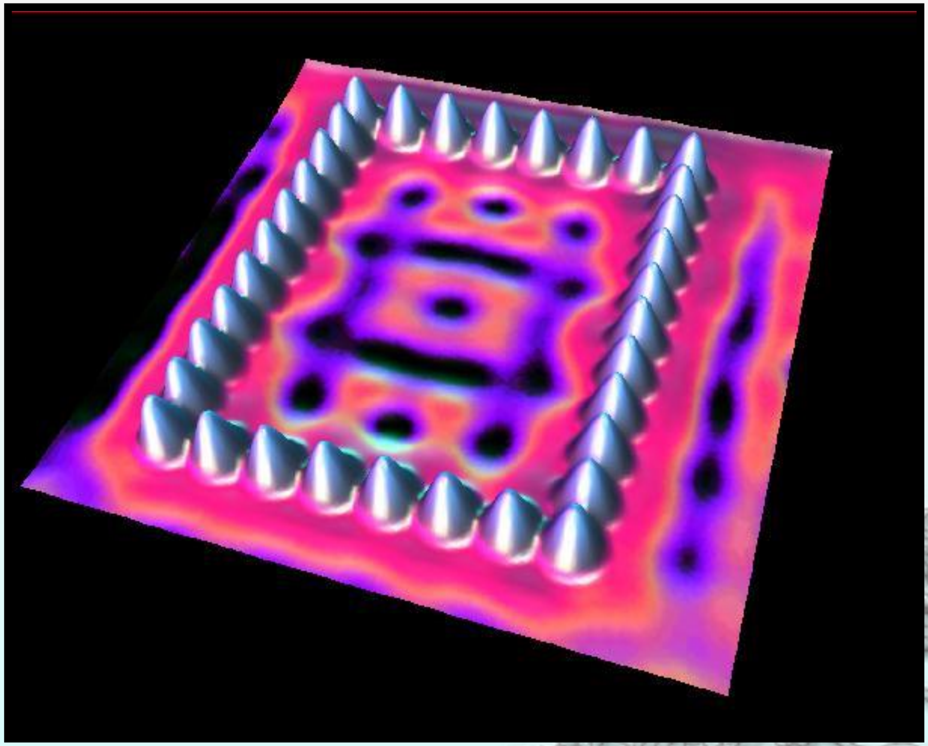
SPM Lithography

- STM can move atoms around on a surface.

Iron on Copper



Iron on Copper



Eigler, et al. from IBM

Comparison of AFM and Other imaging techniques

- AFM vs. STM
  - resolution of STM is better than that of AFM
  - STM is applicable only to conducting samples
  - AFM is applicable to both conductors and insulators
- AFM vs. SEM
  - AFM provides extraordinary topographic contrast direct height measurements and unobscured views of surface features
  - No (conductive) coating is necessary for AFM samples
- AFM vs. TEM
  - 3D AFM image obtained without expensive sample preparation
  - More complete information than two dimensional profiles from TEM

Portland State University

Nanoelectronics & Packaging Lab

AFM vs STM

- Resolution of STM is better than AFM because of the exponential dependence of the tunneling current on distance.

- STM is generally applicable only to conducting samples while AFM is applied to both conductors and insulators.

- AFM offers the advantage that the writing voltage and tip-to-substrate spacing can be controlled independently, whereas with STM the two parameters are integrally linked.

Summary

STM

- real space imaging
- high lateral and vertical resolution
- STS probe electronic properties
- sensitive to noise
- image quality depends on tip conditions
- not true topographic imaging
- only for conductive materials

AFM and others

- apply to nonconducting materials : biomolecules, ceramic
- real topographic imaging
- probe various physical properties: magnetic, electrostatic, hydrophobicity, friction, elastic modulus, etc.
- can manipulate molecules and fabricate nanostructures
- low lateral resolution ~50A
- contact mode can damage the sample
- image distortion due to the presence of water
- difficulties in chemical identification

Video demo

applications